

Exercises

Level - 1

(Problems Based on Fundamentals)

ABC OF HYPERBOLA

1. Find the centre, the vertices, the co-vertices, the length of transverse axis, the conjugate axis and the latus rectum, the eccentricity, the foci and the equation of directrices of each of the following hyperbolas.
 - (i) $\frac{x^2}{9} - \frac{y^2}{4} = 1$
 - (ii) $\frac{x^2}{16} - \frac{y^2}{9} = -1$
 - (iii) $9x^2 - 16y^2 - 36x + 96y - 252 = 0$
2. Find the equation of the hyperbola, whose centre is (1, 0), one focus is (6, 0) and the length of transverse axis is 6.
3. Find the equation of the hyperbola, whose centre is (3, 2) one focus is (5, 2) and one vertex is (4, 2).
4. Find the equation of the hyperbola, whose centre is (-3, 2), one vertex is (-3, 4) and eccentricity is $5/2$.
5. Find the equation of the hyperbola, whose one focus is (2, 1), the directrix is $x + 2y = 1$ and the eccentricity is 2.
6. Find the equation of the hyperbola, whose distance between foci is 16 and the eccentricity is $\sqrt{2}$.
7. Find the equation of the hyperbola, whose foci are (6, 4) and (-6, 4) and the eccentricity is 2.
8. Find the eccentricity of the hyperbola whose latus rectum is half of its transverse axis.
9. If e_1 and e_2 be the eccentricities of a hyperbola and its conjugate, prove that $\frac{1}{e_1^2} + \frac{1}{e_2^2} = 1$.
10. An ellipse and a hyperbola are confocal (have the same focus) and the conjugate axis of the hyperbola is equal to the minor axis of the ellipse. If e_1 and e_2 are the eccentricities of the ellipse and the hyperbola, prove that $e_1^2 + e_2^2 = 2$.
11. Find the centre of the hyperbola

$$4(2y - x - 3)^2 - 9(2x + y - 1)^2 = 80$$
12. Find the centre of the hyperbola

$$3x^2 - 5y^2 - 6x + 20y - 32 = 0$$

13. Prove that the straight lines

$$\frac{x}{a} - \frac{y}{b} = 2013 \text{ and } \frac{x}{a} + \frac{y}{b} = \frac{1}{2013}$$

always meet on a hyperbola.

14. Prove that the locus represented by $x = 3\left(\frac{1+t^2}{1-t^2}\right)$ and $y = \frac{4t}{t^2-1}$ is a hyperbola.

15. Prove that the locus represented by $x = \frac{1}{2}(e^t + e^{-t})$ and $y = \frac{1}{2}(e^t - e^{-t})$ is a hyperbola.

16. If the equation $\frac{x^2}{2014-\lambda} + \frac{y^2}{2013-\lambda} = 1$ represents a hyperbola, find λ .

17. If the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola

$$\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$$
 coincide, find the value of b^2 .

18. If the latus rectum subtends right angle at the centre of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, find its eccentricity.

19. Find the location of point (1, 4) w.r.t the hyperbola $2x^2 - 3y^2 = 6$.

20. If $(\lambda, -1)$ is an exterior point of the curve $4x^2 - 3y^2 = 1$ such that the length of the interval where λ lies is m , find the value of $m + 10$.

INTERSECTIONS OF A LINE AND A HYPERBOLA

21. Find the points common to the hyperbola $25x^2 - 9y^2 = 225$ and the straight line $25x + 12y - 45 = 0$.

22. For what value of λ , does the line $y = 3x + \lambda$ touch the hyperbola $9x^2 - 5y^2 = 45$?

23. For all real values of m , the straight line $y = mx + \sqrt{9m^2 - 4}$ is a tangent to a hyperbola, find the equation of the hyperbola.

24. Find the equations of tangents to the curve $4x^2 - 9y^2 = 36$, which is parallel to $5x - 4y + 7 = 0$.

25. Find the equations of tangents to the curve $9x^2 - 16y^2 = 144$, which is perpendicular to the straight line $3x + 4y + 10 = 0$.

26. If the line $5x + 12y - 9 = 0$ touches the hyperbola $x^2 - 9y^2 = 9$, then find its point of contact.

27. Find the equation of tangents to the curve $4x^2 - 9y^2 = 36$ from the point (3, 2).

28. Find the number of tangents from the point (1, -2) to the curve $2x^2 - 3y^2 = 12$.

29. Find the equation of the tangent to the curve $3x^2 - 4y^2 = 12$ having slope 4.

TANGENT AND TANGENCY

30. Find the equation of the tangent to the curve $x^2 - y^2 - 8x + 2y + 11 = 0$ at (2, 1).

31. Find the equation of the tangent to the curve $4x^2 - 3y^2 = 24$ at $y = 2$.

32. Find the angle between the tangents to the curve $9x^2 - 16y^2 = 144$ drawn from the point (4, 3).

33. Find the equations of the common tangent to the curves $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{x^2}{b^2} - \frac{y^2}{a^2} = -1$.

34. Find the equations of the common tangents to the curves $\frac{x^2}{9} - \frac{y^2}{4} = 1$ and $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

35. Find the equation of the common tangents to the curves $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and $x^2 + y^2 = 9$.

36. Find the equation of the common tangents to the curves $y^2 = 8x$ and $\frac{x^2}{9} - \frac{y^2}{5} = 1$.

37. Find the locus of the point of intersection of the perpendicular tangents to the curve $\frac{x^2}{16} - \frac{y^2}{9} = 1$.

38. Find the product of the perpendiculars from foci upon any tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

39. If the tangent to the parabola $y^2 = 4ax$ intersects the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at A and B respectively, find the locus of the points of intersection of tangents at A and B .

NORMAL AND NORMALCY

40. Find the equation of the normal to the curve $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at $(8, 3\sqrt{3})$.

41. A normal is drawn at one end of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ which meets the axes at points A and B respectively. Find the area of the ΔOAB .

42. Prove that the locus of the foot of the perpendicular from the centre upon any normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$(x^2 + y^2)(a^2y^2 - b^2x^2) = (a^2 + b^2)x^2y^2$$

43. A normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the axes in M and N and the lines MP and NP are drawn perpendiculars to the axes meeting at P . Prove that the locus of P is the hyperbola $a^2x^2 - b^2y^2 = (a^2 + b^2)^2$.

44. If the normal at ϕ on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meets the transverse axis at G such that

$$AG \cdot A'G = a^m(e^n \sec^p \theta - 1),$$

where A, A' are the vertices of the hyperbola and m, n and p are positive integers, find the value of $(m + n + p)^2 + 36$.

45. If the normals at $(x_i, y_i), i = 1, 2, 3, 4$ on the rectangular hyperbola $xy = c^2$ meet at the point (α, β) , prove that

$$(i) \quad x_1 + x_2 + x_3 + x_4 = \alpha$$

$$(ii) \quad y_1 + y_2 + y_3 + y_4 = \beta$$

$$(iii) \quad x_1^2 + x_2^2 + x_3^2 + x_4^2 = \alpha^2$$

$$(iv) \quad y_1^2 + y_2^2 + y_3^2 + y_4^2 = \beta^2$$

$$(v) \quad x_1 \cdot x_2 \cdot x_3 \cdot x_4 = -c^4$$

$$(vi) \quad y_1 \cdot y_2 \cdot y_3 \cdot y_4 = -c^4$$

46. If the normals at $(x_i, y_i), i = 1, 2, 3, 4$ on the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ are concurrent, prove that}$$

$$(i) \quad (x_1 + x_2 + x_3 + x_4) \left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} \right) = 4$$

$$(ii) \quad (y_1 + y_2 + y_3 + y_4) \left(\frac{1}{y_1} + \frac{1}{y_2} + \frac{1}{y_3} + \frac{1}{y_4} \right) = 4$$

CHORD OF CONTACT/CHORD BISECTED AT A POINT

47. Find the equation of the chord of contact of tangents from the point $(2, 3)$ to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

48. Find the locus of the mid-points of the portions of the tangents to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ included between the axes.

49. From the points on the circle $x^2 + y^2 = a^2$, tangents are drawn to the hyperbola $x^2 - y^2 = a^2$. Prove that the locus of the mid-points of the chord of contact is

$$(x^2 - y^2)^2 = a^2(x^2 + y^2)$$

50. Prove that the locus of the mid-points of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ which subtend right angle at the centre is

$$\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} \right)^2 \left(\frac{1}{a^2} - \frac{1}{b^2} \right) = \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} \right)$$

51. Tangents are drawn from a point P to the parabola $y^2 = 4ax$. If the chord of contact of the parabola be a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, find the locus of the point P .

52. A tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cuts the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in points P and Q . Find the locus of the mid-point PQ .

53. Chords of the hyperbola $x^2 - y^2 = a^2$ touch the parabola $y^2 = 4ax$. Prove that the locus of their mid-points is the curve $y^2(x - a) = x^3$.

54. A variable chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is a tangent to the circle $x^2 + y^2 = c^2$. Prove that the locus of its mid-points is

$$\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} \right)^2 = c^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} \right)$$

55. A variable chord of the circle $x^2 + y^2 = a^2$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Prove that the locus of its mid-points is $(x^2 + y^2)^2 = a^2x^2 - b^2y^2$.

56. A tangent to the parabola $y^2 = 4ax$ meets the hyperbola $xy = c^2$ in two points P and Q . Prove that the locus of the mid-point of PQ lies on a parabola.

57. From a point P , tangents are drawn to the circle $x^2 + y^2 = a^2$. If the chord of contact of the circle is a normal chord of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, prove that the locus

$$\text{of the point } P \text{ is } \left(\frac{a^2}{x^2} - \frac{b^2}{y^2} \right) = \left(\frac{a^2 + b^2}{a^2} \right)^2.$$

58. Prove that the locus of the mid-points of the focal chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{ex}{a}$.

59. If the chords of contact of tangents from two points (x_1, y_1) and (x_2, y_2) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are

$$\text{at right angles such that } \frac{x_1 x_2}{y_1 y_2} = -\frac{a^m}{b^n}, \text{ where } m, n \text{ are}$$

$$\text{positive integers, find the value of } \left(\frac{m+n}{4} \right)^{10}.$$

POLE AND POLAR

60. Find the polar of the focus $(-ae, 0)$ with respect to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

61. If the polars of (x_1, y_1) and (x_2, y_2) with respect to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are at right angles, prove that

$$\frac{x_1 x_2}{y_1 y_2} + \frac{a^4}{b^4} = 0.$$

62. Find the pole of the line $x - y = 3$ w.r.t. the hyperbola $x^2 - 3y^2 = 3$.

63. Prove that the locus of the poles of the normal chords with respect to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is the curve $y^2a^6 - x^2b^6 = (a^2 + b^2)2x^2y^2$.

64. Prove that the locus of the poles with respect to the parabola $y^2 = 4ax$ of the tangent to the hyperbola $x^2 - y^2 = a^2$ is the ellipse $4x^2 + y^2 = 4a^2$.

65. Prove that the locus of the pole with respect to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ of any tangent to the circle, whose diameter is the line joining the foci, is the ellipse $\frac{x^2}{a^4} + \frac{y^2}{b^4} = \frac{1}{a^2 + b^2}$.

DIAMETER

66. Prove that the equation of the diameter which bisects the chord $7x + y - 2 = 0$ of the hyperbola $\frac{x^2}{3} - \frac{y^2}{7} = 1$ is $x + 3y = 0$.

67. Find the equation of the diameter of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$, which corresponds the line $3x + 4y + 10 = 0$.

68. Find the equation of the diameter to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ parallel to the chord $2x + 3y + 5 = 0$.

69. In the hyperbola $16x^2 - 9y^2 = 144$, find the equation of the diameter which is conjugate to the diameter whose equation is $x = 2y$.

ASYMPTOTES

70. Find the asymptotes of the curve $xy - 3y - 2x = 0$.

71. Find the equations of the asymptotes of the curve $(a \sec \phi, a \tan \phi)$.

72. Find the eccentricity of the hyperbola whose asymptotes are $3x + 4y = 10$ and $4x - 3y = 5$.

73. Find the equation of a hyperbola whose asymptotes are $2x - y = 3$ and $3x + y = 7$ and which pass through the point $(1, 1)$.

74. The asymptotes of a hyperbola having centre at the point $(1, 2)$ are parallel to the lines $2x + 3y = 0$ and $3x + 2y = 0$. If the hyperbola passes through the point $(5, 3)$, prove that its equation is $(2x + 3y - 8)(3x + 2y - 7) - 154 = 0$.

75. Find the product of the lengths of the perpendiculars from any point on the hyperbola $x^2 - 2y^2 = 2$ to its asymptotes.

76. Find the area of the triangle formed by any tangent to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ and its asymptotes.

77. Let P be a variable point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

such that its distance from the transverse axis is equal to its distance from an asymptote to the given hyperbola. Prove that the locus of P is $(x^2 - y^2)^2 = 4x^2(x^2 - a^2)$.

78. Show that the tangent at any point of a hyperbola cuts off a triangle of constant area from the asymptotes and that the portion of it intercepted between the asymptotes is bisected at the point of contact.

79. If p_1 and p_2 are the perpendiculars from any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ on its asymptotes, prove that $\frac{1}{p_1 p_2} = \frac{1}{a^2} + \frac{1}{b^2}$.

80. If the normal at t_1 to the hyperbola $xy = c^2$ meets it again at t_2 , prove that $t_1^3 t_2 = -1$.

81. A triangle has its vertices on a rectangular hyperbola. Prove that the orthocentre of the triangle also lies on the same hyperbola.

82. Find the locus of the poles of the normal chords of the rectangular hyperbola $xy = c^2$.

83. If the angle between the asymptote is 2α , prove that the eccentricity of the hyperbola is $\sec \alpha$.

84. A circle cuts the rectangular hyperbola $xy = 1$ in points (x_r, y_r) , $r = 1, 2, 3, 4$, prove that $x_1 x_2 x_3 x_4 = 1$ and $y_1 y_2 y_3 y_4 = 1$.

85. If the tangent and the normal to a rectangular hyperbola $xy = c^2$ at a point cuts off intercepts a_1 and a_2 on one axis and b_1, b_2 on the other axis, prove that $a_1 a_2 + b_1 b_2 = 0$.

86. If e_1 and e_2 be the eccentricities of the hyperbola $xy = c^2$ and $x^2 - y^2 = a^2$, find the value of $(e_1 + e_2)^2$.

87. Find the product of the lengths of the perpendiculars

drawn from any point on the hyperbola $\frac{x^2}{2} - y^2 = 1$ to its asymptote.

88. If A, B and C be three points on the rectangular hyperbola $xy = c^2$, find

(i) the area of the $\triangle ABC$

(ii) the area of the triangle formed by the tangents at A, B and C .

89. Find the length of the transverse axis of the rectangular hyperbola $xy = 18$.

90. Prove that the locus of a point whose chord of contact with respect to the circle $x^2 + y^2 = 4$ is a tangent to the hyperbola $xy = 1$ is a hyperbola.

91. Find the asymptotes of the hyperbola $xy = hx + ky$.

92. If e be the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and θ is the angle between the asymptotes, find $\cos(\theta/2)$.

93. A ray is emanating from the point $(5, 0)$ is incident on the hyperbola $9x^2 - 16y^2 = 144$ at a point P with abscissa 8. Find the equation of the reflected ray after first reflection and point P lies in first quadrant.

Hints & Solutions

Level - 1

1. (i) The equation of the given hyperbola is

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

(a) Centre : (0, 0)

(b) Vertices : A(a, 0) = A(3, 0)
and A(-a, 0) = A(-3, 0)(c) Co-vertices : B(0, b) = B(0, 2)
and B'(0, -b) = B'(0, -2)

(d) The length of the transverse axis = 2a = 6

(e) The length of the conjugate axis = 2b = 4

(f) The length of the latus rectum = $\frac{2b^2}{a} = \frac{8}{3}$ (g) Eccentricity = $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{4}{9}} = \frac{\sqrt{13}}{3}$

(h) Equation of the directrices :

$$x = \pm \frac{a}{e} = \pm \frac{2}{\sqrt{13}/3} = \pm \frac{6}{\sqrt{13}}$$

- (ii) The equation of the given hyperbola is

$$\frac{x^2}{16} - \frac{y^2}{9} = -1$$

(a) Centre : (0, 1)

(b) Vertices : A(0, b) = A(0, 3)
and A(0, -b) = A(0, -3)(c) Co-vertices : B(a, 0) = B(4, 0)
and B'(-a, 0) = B'(-4, 0)

(d) The length of the transverse axis = 2a = 6

(e) The length of the conjugate axis = 2b = 8

(f) The length of the latus rectum = $\frac{2a^2}{b} = \frac{16}{3}$ (g) Eccentricity = $e = \sqrt{1 + \frac{a^2}{b^2}} = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$

(h) Equation of the directrices :

$$y = \pm \frac{b}{e} = \pm \frac{2}{5/4} = \pm \frac{6}{5}$$

- (iii) The equation of the hyperbola is

$$9x^2 - 16y^2 - 36x + 96y - 252 = 0$$

$$\Rightarrow 9(x^2 - 4x) - 16(y^2 - 6y) = 252$$

$$\Rightarrow 9(x-2)^2 - 16(y-3)^2 = 252 + 36 - 144 = 144$$

$$\Rightarrow \frac{9(x-2)^2}{144} - \frac{16(y-3)^2}{144} = 1$$

$$\Rightarrow \frac{(x-2)^2}{16} - \frac{(y-3)^2}{9} = 1$$

(a) Centre : (0, 0)

$$\Rightarrow X = 0, Y = 0$$

$$\Rightarrow x - 2 = 0, y - 3 = 0$$

$$\Rightarrow x = 2 \text{ and } y = 3$$

Hence, the centre is (2, 3).

(b) Vertices : $(\pm a, 0)$

$$\Rightarrow X = \pm a, Y = 0$$

$$\Rightarrow x - 2 = 4, y - 3 = 0$$

$$\Rightarrow x = 2 \pm 4, y = 3$$

Hence, the vertices are $(6, 3)$ and $(-2, 3)$.

(c) Co-vertices: $(0, \pm b)$

$$\Rightarrow X = 0, Y = \pm b$$

$$\Rightarrow x - 2 = 0, y - 3 = \pm 3$$

$$\Rightarrow x = 2, y = 3 \pm 3$$

Hence, the co-vertices are $(2, 6)$ and $(2, 0)$

(d) The length of the transverse axis $= 2a = 8$

(e) The length of the conjugate axis $= 2b = 6$

(f) Eccentricity $= e = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$

(g) Co-ordinates of Foci : $(\pm ae, 0)$

$$X = \pm 5, Y = 0$$

$$\Rightarrow X - 2 = \pm 5, y - 3 = 0$$

$$\Rightarrow X = 2 \pm 5, y = 3$$

Hence, the co-ordinates of the foci are $(7, 3)$ and $(-3, 3)$

2. The equation of the hyperbola with centre $(1, 0)$ is

$$\frac{(x-1)^2}{a^2} - \frac{y^2}{b^2} = 1$$

Here, $2a = 6 \Rightarrow a = 3$

Also, one focus $= (6, 0)$

$$\Rightarrow 1 + ae = 6$$

$$\Rightarrow ae = 5$$

$$\Rightarrow 3e = 5$$

$$\Rightarrow e = 5/3$$

Therefore,

$$b^2 = a^2(e^2 - 1) = 9\left(\frac{25}{9} - 1\right) = 16$$

Hence, the equation of the hyperbola is

$$\frac{(x-1)^2}{9} - \frac{y^2}{16} = 1$$

3. Since the focus is $(5, 2)$ and the vertex is $(4, 2)$, so the axis of the hyperbola is parallel to x -axis and $a = 1$, $ae = 2$

Let the equation of the hyperbola be

$$\frac{(x-3)^2}{a^2} - \frac{(y-2)^2}{b^2} = 1 \quad \dots(i)$$

As we know that, the relation in a, b and e with respect to a hyperbola is

$$b^2 - a^2(e^2 - 1) - a^2e^2 - a^2 - 4 - 1 - 3$$

Now from Eq. (i), we get

$$\frac{(x-3)^2}{1} - \frac{(y-2)^2}{3} = 1$$

4. Since the focus is $(-3, 2)$ and the vertex is $(-3, 4)$, so the axis of the hyperbola is parallel to y -axis and $be = 2$

$$\Rightarrow b \times \frac{5}{2} = 2$$

$$\Rightarrow b = \frac{4}{5}$$

$$\text{Also, } a^2 = b^2(e^2 - 1) = \frac{16}{25}\left(\frac{25}{4} - 1\right) = \frac{84}{25}$$

Hence, the equation of the hyperbola is

$$\frac{(x+3)^2}{a^2} - \frac{(y-2)^2}{b^2} = 1$$

$$\Rightarrow \frac{(x+3)^2}{84} - \frac{(y-2)^2}{16} = \frac{1}{25}$$

5. From the definition of the hyperbola, we can write

$$\frac{SP}{PM} = e,$$

where S = focus, $P = (x, y)$

$$\Rightarrow SP^2 = e^2 PM^2$$

$$\Rightarrow (x-2)^2 + (y-1)^2 = 4\left\{\left(\frac{x+2y-1}{\sqrt{1+4}}\right)^2\right\}$$

$$\Rightarrow 5\{(x-2)^2 + (y-1)^2\} = 4(x+2y-1)^2$$

6. Given relation is

$$2ae = 16$$

$$\Rightarrow ae = 8$$

$$\Rightarrow a = 8/e \Rightarrow a = 2$$

$$\text{Now, } b^2 - a^2(e^2 - 1) - a^2e^2 - a^2 = 64 - 4 = 60$$

Hence, the equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{x^2}{4} - \frac{y^2}{60} = 1$$

7. Since foci are $(6, 4)$ and $(-6, 4)$, so the axis of the hyperbola is parallel to x -axis.

It is given that the distance between two foci is 12, so

$$2ae = 12$$

$$\Rightarrow ae = 6$$

$$\Rightarrow 2a = 6$$

$$\Rightarrow a = 3$$

Also,

$$b^2 - a^2(e^2 - 1) - a^2e^2 - a^2 - 36 - 9 = 27$$

Hence, the equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{27} = 1$$

8. Given relation is

$$\frac{2b^2}{a} = \frac{1}{2} \times (2a) = a$$

$$\Rightarrow \frac{2b^2}{a} = a$$

$$\Rightarrow 2b^2 = a^2$$

Also, relation in a, b and e is

$$b^2 = a^2(e^2 - 1)$$

$$\Rightarrow \frac{a^2}{2} = a^2(e^2 - 1)$$

$$\Rightarrow (e^2 - 1) = \frac{1}{2}$$

$$\Rightarrow e = \sqrt{\frac{3}{2}}$$

9. Let the equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and

its conjugate is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$.

Therefore,

$$e_1 = \sqrt{1 + \frac{b^2}{a^2}} \text{ and } e_2 = \sqrt{1 + \frac{a^2}{b^2}}$$

$$\Rightarrow e_1^2 = \left(\frac{a^2 + b^2}{a^2} \right) \text{ and } e_2^2 = \left(\frac{a^2 + b^2}{b^2} \right)$$

$$\Rightarrow \frac{1}{e_1^2} + \frac{1}{e_2^2} = \left(\frac{a^2}{a^2 + b^2} \right) + \left(\frac{b^2}{a^2 + b^2} \right) = \left(\frac{a^2 + b^2}{a^2 + b^2} \right) = 1$$

10. Let the equation of the ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the

equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Also, it is given that,

$$2b = 2a$$

$$\Rightarrow b = a$$

Let e_1 and e_2 be the eccentricities of the ellipse and the hyperbola.

$$\text{Then } e_1 = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{a^2}{a^2}} = 0$$

$$\text{and } e_2 = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{a^2}{a^2}} = \sqrt{2}$$

$$\text{Thus, } e_1^2 + e_2^2 = 2$$

11. The equation of the hyperbola is

$$4(2y - x - 3)^2 - 9(2x + y - 1)^2 = 80$$

The centre of the hyperbola is obtained from the equations

$$2y - x - 3 = 0 \text{ and } 2x + y - 1 = 0$$

Solving, we get,

$$x = -2/5$$

$$\text{and } y = 13/10$$

$$\text{Hence, the centre is } \left(-\frac{2}{5}, \frac{13}{10} \right).$$

12. The equation of the given hyperbola is

$$3x^2 - 5y^2 - 6x + 20y - 32 = 0$$

$$\Rightarrow 3(x^2 - 2x) - 5(y^2 - 4y) = 32$$

$$\Rightarrow 3(x - 1)^2 - 5(y - 2)^2 = 32 + 3 - 20 = 15$$

$$\Rightarrow \frac{3(x - 1)^2}{15} - \frac{5(y - 2)^2}{15} = 1$$

$$\Rightarrow \frac{(x - 1)^2}{5} - \frac{(y - 2)^2}{3} = 1$$

13. The given straight lines are

$$\frac{x}{a} - \frac{y}{b} = 2013 \quad \dots(i)$$

and

$$\frac{x}{a} + \frac{y}{b} = \frac{1}{2013} \quad \dots(ii)$$

Multiplying Eqs (i) and (ii), we get

$$\left(\frac{x}{a} - \frac{y}{b} \right) \left(\frac{x}{a} + \frac{y}{b} \right) = 2013 \times \frac{1}{2013} = 1$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

which represents a hyperbola.

14. We have,

$$x = 3 \left(\frac{1 + t^2}{1 - t^2} \right) \text{ and } y = \frac{4t}{t^2 - 1}$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{1 + t^2}{1 - t^2} \right)^2 - \left(\frac{2t}{1 - t^2} \right)^2$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = \frac{(1 + t^2)^2 - 4t^2}{(1 - t^2)^2} = \left(\frac{1 - t^2}{1 - t^2} \right)^2 = 1$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = 1$$

which represents a hyperbola.

15. We have,

$$x = \frac{1}{2}(e^t + e^{-t}) \text{ and } y = \frac{1}{2}(e^t - e^{-t})$$

$$\Rightarrow 2x = (e^t + e^{-t}) \text{ and } 2y = (e^t - e^{-t})$$

$$\Rightarrow 4x^2 - 4y^2 = (e^t + e^{-t})^2 - (e^t - e^{-t})^2$$

$$\Rightarrow 4x^2 - 4y^2 = 2 + 2$$

$$\Rightarrow x^2 - y^2 = 1$$

Which represents a rectangular hyperbola.

16. The given equation of the hyperbola is

$$\frac{x^2}{2014 - \lambda} + \frac{y^2}{2013 - \lambda} = 1$$

$$\Rightarrow (2013 - \lambda)x^2 + (2014 - \lambda)y^2 - (2013 - \lambda)(2014 - \lambda) = 0$$

$$\Rightarrow (2013 - \lambda)x^2 + (\lambda - 2014)y^2 - (2013 - \lambda)(2014 - \lambda) = 0$$

The given equation represents a hyperbola, if

$$h^2 - ab > 0$$

$$\Rightarrow 0 - (2013 - \lambda)(\lambda - 2014) > 0$$

$$\Rightarrow (2013 - \lambda)(\lambda - 2014) < 0$$

$$\Rightarrow 2013 < \lambda < 2014$$

$$\Rightarrow \lambda \in (2013, 2014)$$

17. The given equation of the hyperbola is

$$\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$$

We have,

$$e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{81}{144}} = \sqrt{\frac{225}{144}} = \frac{15}{12} = \frac{5}{4}$$

Also, $a^2 = \frac{144}{25}$

Thus, the foci are

$$(\pm ae, 0) = \left(\pm \frac{12}{5} \times \frac{5}{4}, 0\right) = (\pm 3, 0).$$

Now, for the ellipse,

$$ae = 3$$

$$\Rightarrow a^2 e^2 = 9$$

$$\text{Thus, } b^2 = a^2(1 - e^2)$$

$$= a^2 - a^2 e^2$$

$$= 16 - 9 = 7$$

Hence, the value of b^2 is 7.

18. Let LL' be the latus rectum of the given hyperbola.

Therefore, $L\left(ae, \frac{b^2}{a}\right)$ and $L'\left(ae, -\frac{b^2}{a}\right)$ and the centre of the hyperbola is $C(0, 0)$

$$\text{Now, slope of } CL = m_1 = \frac{(b^2/a)}{ae} = \frac{b^2}{a^2 e}$$

$$\text{and the slope of } m_2 = \frac{(-b^2/a)}{ae} = -\frac{b^2}{a^2 e}$$

Since, the latus rectum subtends right angle at the centre, so

$$m_1 \times m_2 = -1$$

$$\Rightarrow \left(\frac{b^2}{a^2 e}\right) \times \left(-\frac{b^2}{a^2 e}\right) = -1$$

$$\Rightarrow \left(\frac{b^4}{a^4 e^2}\right) = 1$$

$$\Rightarrow b^4 = a^4 e^2$$

$$\Rightarrow a^4(e^2 - 1)^2 = a^4 e^2$$

$$\Rightarrow (e^2 - 1)^2 = e^2$$

$$\Rightarrow e^4 - 3e^2 + 1 = 0$$

$$\Rightarrow e^2 = \frac{3 \pm \sqrt{5}}{2}$$

$$\Rightarrow e^2 = \frac{3 + \sqrt{5}}{2}$$

$$\Rightarrow e^2 = \frac{3 + \sqrt{5}}{2} = \frac{6 + 2\sqrt{5}}{4} = \left(\frac{\sqrt{5} + 1}{2}\right)^2$$

$$\Rightarrow e = \left(\frac{\sqrt{5} + 1}{2}\right)$$

19. We have,

$$2x_1^2 - 3y_1^2 - 1 = 2 - 48 - 1 = -47 < 0$$

Thus, the point $(1, 4)$ lies outside of the hyperbola.

20. Since the point $(\lambda, -1)$ is an exterior point of the curve

$$4x^2 - 3y^2 = 1, \text{ so}$$

$$4\lambda^2 - 3 - 1 < 0$$

$$\Rightarrow 4\lambda^2 - 4 < 0$$

$$\Rightarrow \lambda^2 - 1 < 0$$

$$\Rightarrow (\lambda + 1)(\lambda - 1) < 0$$

$$\Rightarrow -1 < \lambda < 1$$

$$\Rightarrow \lambda \in (-1, 1)$$

Thus, the length of the interval, where λ lies is 2.

Therefore, $m = 2$

Hence, the value of $m + 10 = 2 + 10 = 12$.

21. As we know that if the line $y = mx + c$ be a tangent

to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the co-ordinates of the

point of contact is $\left(\pm \frac{a^2 m}{c}, \pm \frac{b^2}{c}\right)$.

The equation of the given hyperbola is

$$25x^2 - 9y^2 = 225$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{25} = 1 \quad \dots(i)$$

Also, the given line is

$$25x + 12y = 45$$

$$\Rightarrow 12y = -25x + 45$$

$$\Rightarrow y = -\frac{25}{12}x + \frac{45}{12} \quad \dots(ii)$$

Here, $a = 3$, $b = 5$ and $m = -25/12$.

Thus, the common point is

$$\left(\pm \frac{a^2 m}{c}, \pm \frac{b^2}{c}\right) = \left(\mp 5, \pm \frac{20}{3}\right).$$

22. As we know that, the line $y = mx + c$ will be a tangent

to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, if

$$c^2 = a^2 m^2 - b^2 \quad \dots(i)$$

The equation of the hyperbola is

$$9x^2 - 5y^2 = 45$$

$$\Rightarrow \frac{x^2}{5} - \frac{y^2}{9} = 1$$

Here, $a^2 = 5$, $b^2 = 9$, $m = 3$, $c = \lambda$

Therefore, from Eq. (i), we get

$$\lambda^2 = a^2 m^2 - b^2 = 15 - 9 = 6$$

$$\Rightarrow \lambda = \pm \sqrt{6}$$

23. The equation of any tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

$$y = mx + \sqrt{a^2 m^2 - b^2} \quad \dots(i)$$

The equation of the given tangent is

$$y = mx + \sqrt{9m^2 - 4} \quad \dots(ii)$$

Since Eqs (i) and (ii) are identical, so

$$a^2 = 9, b^2 = 4$$

Hence, the equation of the hyperbola is

$$\frac{x^2}{9} - \frac{y^2}{4} = 1$$

$$\Rightarrow 4x^2 - 9y^2 = 36$$

24. The equation of any tangent to the parallel to $5x - 4y + 7 = 0$ is

$$5x - 4y + \lambda = 0$$

$$\Rightarrow 4y = 5x + \lambda$$

$$\Rightarrow y = \frac{5}{4}x + \frac{\lambda}{4} \quad \dots(i)$$

The equation of the given hyperbola is

$$4x^2 - 9y^2 = 36$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = 1 \quad \dots(ii)$$

Since, the line (i) is a tangent to the hyperbola (ii), so

$$\frac{\lambda^2}{16} = 9 \left(\frac{25}{16} \right) - 4$$

$$\Rightarrow \frac{\lambda^2}{16} = \frac{225 - 64}{16}$$

$$\Rightarrow \lambda^2 = 161$$

$$\Rightarrow \lambda = \pm\sqrt{161}$$

Hence, the equations of the tangents are

$$5x - 4y \pm \sqrt{161} = 0$$

25. The equation of any line perpendicular to

$$3x + 4y + 10 = 0 \text{ is}$$

$$4x - 3y + \lambda = 0$$

$$\Rightarrow y = \left(\frac{4}{3} \right)x + \frac{\lambda}{3} \quad \dots(i)$$

The equation of the given hyperbola is

$$9x^2 - 16y^2 = 144$$

$$\Rightarrow \frac{x^2}{16} - \frac{y^2}{9} = 1 \quad \dots(ii)$$

The line (i) will be a tangent to the hyperbola (ii), if

$$\left(\frac{\lambda}{3} \right)^2 = 16 \left(\frac{4}{3} \right)^2 - 9$$

$$\Rightarrow \lambda^2 = 256 - 81 = 175$$

$$\Rightarrow \lambda = \pm\sqrt{175} = \pm 5\sqrt{7}$$

Hence, the equation of the tangents are

$$4x - 3y \pm 5\sqrt{7} = 0.$$

26. The given hyperbola is

$$x^2 - 9y^2 = 9$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{1} = 1 \quad \dots(i)$$

The given line is $5x + 12y - 9 = 0$

$$\Rightarrow y = \left(-\frac{5}{12} \right)x + \frac{3}{4} \quad \dots(ii)$$

If the line (ii) be a tangent to the hyperbola (i), the co-ordinates of the point of contact can be

$$\left(\pm \frac{a^2 m}{c}, \pm \frac{b^2}{c} \right) = \left(\pm (-5), \pm \frac{4}{3} \right)$$

27. The equation of the given hyperbola is

$$4x^2 - 9y^2 = 36$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = 1$$

Here, $a^2 = 9, b^2 = 4$.

The equation of any tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

$$y = mx + \sqrt{a^2 m^2 - b^2}$$

$$\Rightarrow y = mx + \sqrt{9m^2 - 4}$$

which is passing through (3, 2). So

$$(2 - 3m)^2 = 9m^2 - 4$$

$$\Rightarrow 4 - 12m + 9m^2 = 9m^2 - 4$$

$$\Rightarrow 4 - 12m = -4$$

$$\Rightarrow m = \frac{2}{3}, m = \infty$$

Hence, the equations of the tangents are

$$x + 3 = 0 \text{ and } y = \frac{2}{3}x.$$

28. We have,

$$2x_1^2 - 3y_1^2 - 12 = 2.1 - 3.4 - 12$$

$$= 2 - 12 - 12 = 2 - 24 = -22 < 0$$

So, the point (1, -2) lies outside of the hyperbola.

Thus, the number of tangents is 2.

29. The equation of the given hyperbola is

$$3x^2 - 4y^2 = 12$$

$$\Rightarrow \frac{x^2}{4} - \frac{y^2}{3} = 1 \quad \dots(i)$$

Here, $a^2 = 4, b^2 = 3$ and $m = 4$

The equation of any tangent to the hyperbola (i) is

$$y = mx + \sqrt{a^2 m^2 - b^2}$$

$$\Rightarrow y = 4x \pm \sqrt{64 - 3}$$

$$\Rightarrow y = 4x \pm \sqrt{61}$$

Hence, the equations of tangents are

$$y = 4x + \sqrt{61} \text{ and } y = 4x - \sqrt{61}.$$

30. The equation of tangent to the curve

$$x^2 - y^2 - 8x + 2y + 11 = 0 \text{ is}$$

$$xx_1 - yy_1 - 4(x - x_1) + (y - y_1) + 11 = 0$$

$$\Rightarrow 2x - y - 4(x + 2) + (y + 1) + 11 = 0$$

$$\Rightarrow -2x + 4 = 0$$

$$\Rightarrow x - 2 = 0$$

31. When $y = 2$, then $4x^2 = 36$

$$\Rightarrow x = \pm 3$$

Hence, the points are $(3, 2)$ and $(-3, 2)$.

The equation of the tangent at $(3, 2)$ is

$$12x - 6y = 24$$

$$\Rightarrow 2x - y = 4$$

Also, the equation of the tangent at $(-3, 2)$ is

$$-12x - 6y = 24$$

$$\Rightarrow 2x + y + 4 = 0$$

32. The equation of the given hyperbola is

$$9x^2 - 16y^2 = 144$$

$$\Rightarrow \frac{x^2}{16} - \frac{y^2}{9} = 1 \quad \dots(i)$$

The equation of any tangent to the hyperbola (i) can be

considered as $y = mx + \sqrt{16m^2 - 9}$

which is passing through $(4, 3)$. So

$$(3 - 4m)^2 = 16m^2 - 9$$

$$\Rightarrow 9 - 24m + 16m^2 = 16m^2 - 9$$

$$\Rightarrow 24m = 18$$

$$\Rightarrow m = \frac{3}{4} \text{ and } m = \infty$$

Let θ be the angle between them. Then

$$\tan(\theta) = \left| \frac{\frac{3}{4} - \infty}{1 + \frac{3}{4} \cdot \infty} \right| = \left| \frac{\frac{3}{4} - 1}{\frac{1}{4} + \frac{3}{4}} \right| = \frac{4}{3}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right)$$

Hence, the angle between the tangents is

$$\tan^{-1}\left(\frac{4}{3}\right).$$

33. The equation of any tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

$$y = m_1 x + \sqrt{a^2 m_1^2 - b^2} \quad \dots(i)$$

The equation of any tangent to the hyperbola

$$\frac{x^2}{(-b^2)} - \frac{y^2}{(-a^2)} = 1 \text{ is}$$

$$y = m_2 x + \sqrt{(-b^2)m_2^2 - (-a^2)} \quad \dots(ii)$$

Since the equations (i) and (ii) are identical, so

$$a^2 m_1^2 - b^2 = (-b^2)m_2^2 - (-a^2)$$

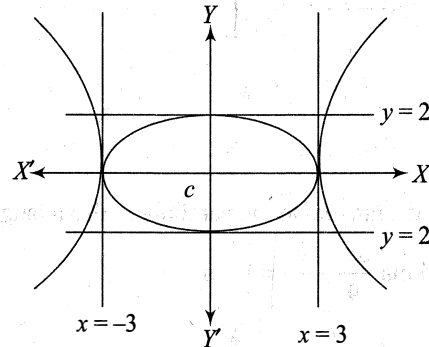
$$\Rightarrow m_1^2 = 1 \text{ and } m_2^2 = 1$$

Thus, $m_1 = \pm 1 = m_2$

Hence, the equations of the common tangents are

$$y = \pm x + \sqrt{a^2 - b^2}$$

34. The equation of the given curves are $\frac{x^2}{9} - \frac{y^2}{4} = 1$ and $\frac{x^2}{9} + \frac{y^2}{4} = 1$.

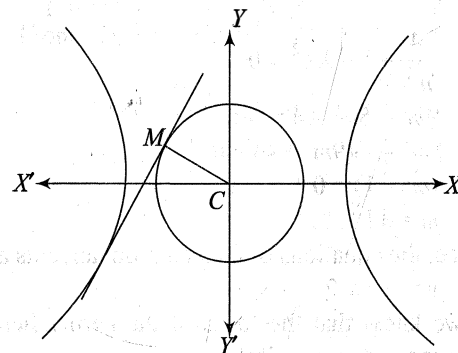


Here, the length of the major axis of the ellipse is equal to the length of the transverse axis and also the length of the minor axis is equal to the length of the conjugate axis. Thus, the equations of the common tangents are $x = \pm 3$ and $y = \pm 2$.

35. The equation of any tangent to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

$$y = mx + \sqrt{16m^2 - 9}$$

$$\Rightarrow mx - y + \sqrt{16m^2 - 9} = 0 \quad \dots(i)$$



If the tangent (i) is also the tangent to the circle $x^2 + y^2 = 9$, the length of the perpendicular from the centre of the circle is equal to the radius of the circle. So

$$\left| \frac{0 - \sqrt{16m^2 - 9}}{\sqrt{m^2 + 1}} \right| = 3$$

$$\Rightarrow 16m^2 - 9 = 9(m^2 + 1)$$

$$\Rightarrow 16m^2 - 9m^2 = 9 + 1 = 10$$

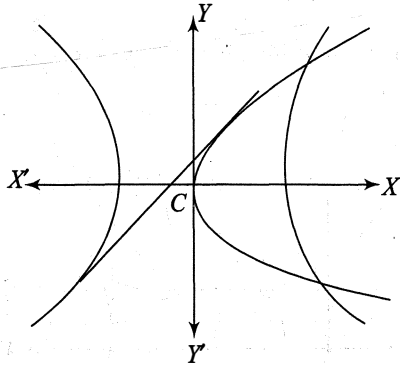
$$\Rightarrow 7m^2 = 10$$

$$\Rightarrow m = \pm \sqrt{\frac{7}{10}}$$

Hence, the equation of tangents are $y = \pm \sqrt{\frac{7}{10}} x + \sqrt{\frac{11}{5}}$.

36. The equation of any tangent to the parabola $y^2 = 8x$ is

$$y = mx + \frac{2}{m} \quad \dots(i)$$



Since the tangent to the parabola is also a tangent to the hyperbola $\frac{x^2}{9} - \frac{y^2}{5} = 1$, so

$$5x^2 - 9\left(mx + \frac{2}{m}\right)^2 = 45$$

$$\Rightarrow 5x^2 - 9\left(m^2x^2 + \frac{4}{m^2} + 4m\right) = 45$$

$$\Rightarrow (5 - 9m^2)x^2 - 36x - 9\left(\frac{4}{m^2} + 5\right) = 0$$

Now,

$$D = 0$$

$$\Rightarrow (36)^2 + 36(5 - 9m^2)\left(\frac{4}{m^2} + 5\right) = 0$$

$$\Rightarrow 36 + \frac{20}{m^2} + 25 - 36 - 45m^2 = 0$$

$$\Rightarrow \frac{4}{m^2} + 5 - 9m^2 = 0$$

$$\Rightarrow 9m^4 - 5m^2 - 4 = 0$$

$$\Rightarrow (m^2 - 1)(9m^2 + 4) = 0$$

$$\Rightarrow (m^2 - 1) = 0$$

$$\Rightarrow m = \pm 1$$

Hence, the equations of the common tangents are $y = \pm x \pm 2$.

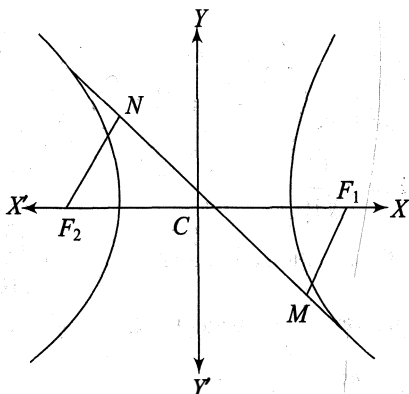
37. As we know that the locus of the perpendicular tangents is the director circle.

Hence, the equation of the director circle is

$$x^2 + y^2 = a^2 - b^2 = 16 - 9 = 7$$

$$\Rightarrow x^2 + y^2 = 7$$

38. Let F_1 and F_2 be two foci of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.



Then $F_1 = (ae, 0)$ and $F_2 = (-ae, 0)$.

The equation of any tangent to the hyperbola is

$$\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1 \quad \dots(i)$$

Let p_1 and p_2 be two perpendiculars from foci upon the tangent (i).

$$\Rightarrow p_1 p_2 = \frac{a^2(e^2 - 1)(e^2 \sec^2 \theta - 1)}{((e^2 - 1) \sec^2 \theta + \tan^2 \theta)}$$

$$\Rightarrow p_1 p_2 = \frac{a^2(e^2 - 1)(e^2 \sec^2 \theta - 1)}{((e^2 - 1) \sec^2 \theta + (\sec^2 \theta - 1))}$$

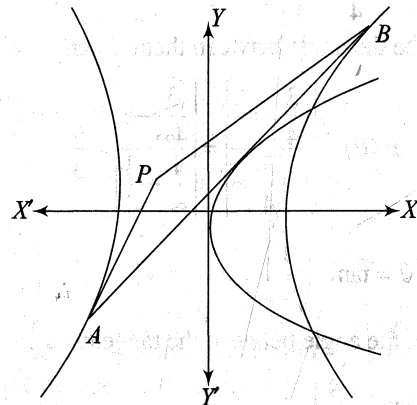
$$\Rightarrow p_1 p_2 = \frac{a^2(e^2 - 1)(e^2 \sec^2 \theta - 1)}{(e^2 \sec^2 \theta - 1)}$$

$$\Rightarrow p_1 p_2 = a^2(e^2 - 1)$$

$$\Rightarrow p_1 p_2 = b^2$$

Hence, the product of the lengths of the perpendiculars is b^2 .

39. Let $P(\alpha, \beta)$ be the point of intersection of tangents at A and B.



Clearly, the point of intersection of the tangents at A and B is the chord of contact.

Therefore, the chord of contact AB is

$$\frac{\alpha x}{a^2} - \frac{\beta y}{b^2} = 1$$

$$\Rightarrow y = \frac{b^2 x \alpha}{a^2 y \beta} - \frac{b^2}{\beta}$$

which is a tangent to the parabola $y^2 = 4ax$. So

$$-\frac{b^2}{\beta} = \frac{a}{\left(\frac{b^2 \alpha}{a^2 \beta}\right)}$$

$$\Rightarrow \beta^2 = \left(-\frac{b^4}{a^3} \alpha\right)$$

Hence, the locus of $P(\alpha, \beta)$ is

$$y^2 = \left(-\frac{b^4}{a^3} x\right)$$

$$\text{Then, } p_1 = \frac{e \sec \theta - 1}{\sqrt{\frac{\sec^2 \theta}{a^2} + \frac{\tan^2 \theta}{b^2}}}$$

$$\text{and } p_2 = \frac{e \sec \theta + 1}{\sqrt{\frac{\sec^2 \theta}{a^2} + \frac{\tan^2 \theta}{b^2}}}$$

$$\begin{aligned} \Rightarrow p_1 p_2 &= \frac{(e \sec \theta - 1)(e \sec \theta + 1)}{\frac{\sec^2 \theta}{a^2} + \frac{\tan^2 \theta}{b^2}} \\ &= \frac{a^2 b^2 (e^2 \sec^2 \theta - 1)}{(b^2 \sec^2 \theta + a^2 \tan^2 \theta)} \\ &= \frac{a^4 (e^2 - 1)(e^2 \sec^2 \theta - 1)}{(a^2 (e^2 - 1) \sec^2 \theta + a^2 \tan^2 \theta)} \end{aligned}$$

40. The equation of the normal to the curve $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at $(8, 3\sqrt{3})$ is

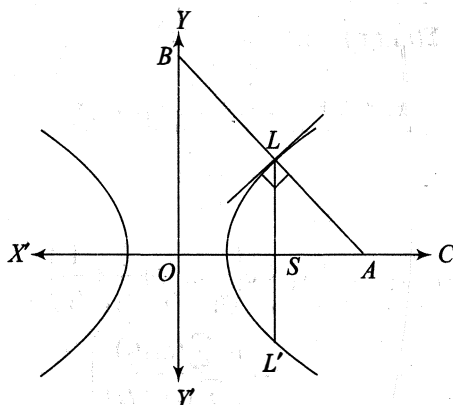
$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$$

$$\Rightarrow \frac{16x}{8} + \frac{9y}{3\sqrt{3}} = 16 + 9$$

$$\Rightarrow 2x + \sqrt{3}y = 25$$

Hence, the required equation of the normal is $2x + \sqrt{3}y = 25$.

41. Let, one end of the latus rectum of the given hyperbola is $L\left(ae, \frac{b^2}{a}\right)$.



The equation of the normal to the given hyperbola at L is

$$\frac{a^2 x}{x_1} + \frac{b^2 y}{y_1} = a^2 + b^2$$

$$\Rightarrow \frac{ax}{e} + ay = a^2 + b^2$$

$$\Rightarrow \frac{x}{\left(\frac{e(a^2 + b^2)}{a}\right)} + \frac{y}{\left(\frac{a^2 + b^2}{a}\right)} = 1$$

Hence, the area of ΔOAB

$$= \frac{1}{2} \times \frac{e(a^2 + b^2)}{a} \times \left(\frac{a^2 + b^2}{a}\right)$$

$$= \frac{1}{2} \times \frac{e(a^2 + b^2)^2}{a^2}$$

$$= \frac{1}{2} \times a^2 e^5$$

42. The equation of any normal at $(a \sec \phi, b \tan \phi)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$ax \cos \phi + by \cot \phi = a^2 + b^2 \quad \dots(i)$$

The equation of any line perpendicular to (i) and passing through the origin is

$$(b \cot \phi)x - (a \cos \phi)y = 0$$

$$\Rightarrow bx - a \sin \phi y = 0 \quad \dots(ii)$$

If we eliminate ϕ between Eqs (i) and (ii), we get the required locus of the foot of the perpendicular.

From (ii), we get,

$$\sin(\phi) = \frac{bx}{ay}$$

$$\Rightarrow \cos(\phi) = \frac{\sqrt{(a^2 y^2 - b^2 x^2)}}{ay}$$

$$\text{and } \cot(\phi) = \frac{\sqrt{(a^2 y^2 - b^2 x^2)}}{bx}$$

From Eq. (i), we get

$$\begin{aligned} ax \times \frac{\sqrt{(a^2 y^2 - b^2 x^2)}}{ay} + by \times \frac{\sqrt{(a^2 y^2 - b^2 x^2)}}{bx} \\ = a^2 + b^2 \end{aligned}$$

$$\Rightarrow (x^2 + y^2) \left(\sqrt{(a^2 y^2 - b^2 x^2)} \right) = (a^2 + b^2) xy$$

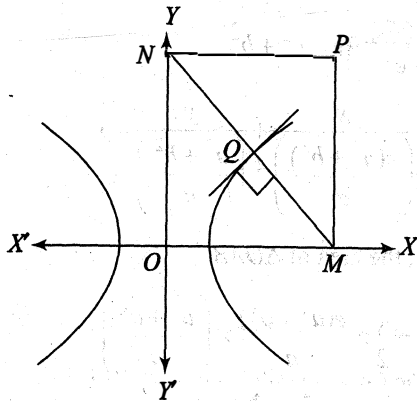
$$\Rightarrow (x^2 + y^2)^2 (a^2 y^2 - b^2 x^2) = (a^2 + b^2)^2 x^2 y^2$$

which is the required locus of the foot of the perpendicular.

43. The equation of any normal to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

$$ax \cos \phi + by \cot \phi = (a^2 + b^2) \quad \dots(i)$$



Since the normal (i) meets the x -axis at M and y -axis at N respectively. Then,

$$M = \left(\frac{a^2 + b^2}{a \cos \phi}, 0 \right) \text{ and } N = \left(0, \left(\frac{a^2 + b^2}{b} \right) \tan \phi \right)$$

Let the co-ordinates of the point P be (α, β) .

Since PM and PN are perpendiculars to the axes, so the co-ordinates of P are

$$\left(\left(\frac{a^2 + b^2}{a} \right) \sec \phi, \left(\frac{a^2 + b^2}{b} \right) \tan \phi \right)$$

Therefore,

$$\alpha = \left(\frac{a^2 + b^2}{a} \right) \sec \phi \text{ and } \beta = \left(\frac{a^2 + b^2}{b} \right) \tan \phi$$

$$\Rightarrow \alpha \left(\frac{a}{a^2 + b^2} \right) = \sec \phi \text{ and } \beta \left(\frac{b}{a^2 + b^2} \right) = \tan \phi$$

As we know that,

$$\sec^2 \phi - \tan^2 \phi = 1$$

$$\alpha^2 \left(\frac{a}{a^2 + b^2} \right)^2 - \beta^2 \left(\frac{b}{a^2 + b^2} \right)^2 = 1$$

$$\Rightarrow \alpha^2 a^2 - \beta^2 b^2 = (a^2 + b^2)^2$$

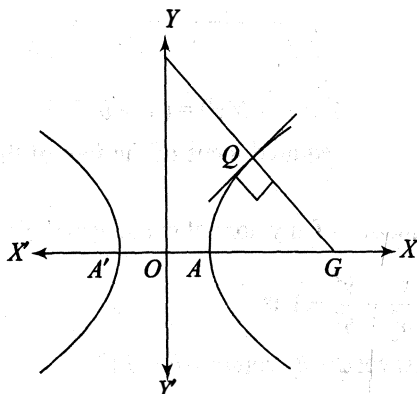
Hence, the locus of (α, β) is

$$\alpha^2 x^2 - \beta^2 y^2 = (a^2 + b^2)^2$$

44. The equation of any normal to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ at } (\phi) \text{ is}$$

$$ax \cos \phi + by \cot \phi = a^2 + b^2$$



$$\text{Thus, the co-ordinates of } G = \left(\frac{a^2 + b^2}{a \cos \phi}, 0 \right)$$

Clearly, the vertices, $A = (a, 0)$ and $A' = (-a, 0)$

$$\text{Now, } AG = \left(\frac{a^2 + b^2}{a \cos \phi} - a \right)$$

$$\text{and } A'G = \left(\frac{a^2 + b^2}{a \cos \phi} + a \right)$$

Therefore,

$$AG \cdot A'G = \left(\frac{a^2 + b^2}{a \cos \phi} - a \right) \left(\frac{a^2 + b^2}{a \cos \phi} + a \right)$$

$$= \left(\left(\frac{a^2 + b^2}{a} \right)^2 \sec^2 \phi - a^2 \right)$$

$$= (a^2 e^2 \sec^2 \phi - a^2)$$

$$= a^2 (e^2 \sec^2 \phi - 1)$$

$$\Rightarrow m = 2, n = 2, p = 2$$

Hence,

$$(m + n + p)^2 + 36 = 36 + 36 = 72.$$

45. The equation of the normal to the hyperbola $xy = c^2$ at

$$\left(ct, \frac{c}{t} \right) \text{ is}$$

$$xt^3 - yt - ct^4 + c = 0$$

$$\Rightarrow ct^4 - xt^3 + yt - c = 0$$

which is passing through (α, β) , so

$$ct^4 - \alpha t^3 + \beta t - c = 0.$$

Let its four roots are t_1, t_2, t_3, t_4 .

$$\text{Therefore, } t_1 + t_2 + t_3 + t_4 = \frac{\alpha}{c},$$

$$\Sigma(t_1 t_2) = 0, \Sigma(t_1 t_2 t_3) = -\frac{\beta}{c}$$

$$\text{and } \Sigma(t_1 t_2 t_3 t_4) = 1.$$

$$(i) x_1 + x_2 + x_3 + x_4 = c(t_1 + t_2 + t_3 + t_4)$$

$$= c \left(\frac{\alpha}{c} \right) = \alpha$$

$$(ii) y_1 + y_2 + y_3 + y_4 = c \left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4} \right)$$

$$= c \left(\frac{\Sigma(t_1 t_2 t_3)}{\Sigma(t_1 t_2 t_3 t_4)} \right)$$

$$= c \left(\frac{\left(-\frac{\beta}{c} \right)}{-1} \right) = \beta$$

$$\begin{aligned} \text{(iii)} \quad x_1^2 + x_2^2 + x_3^2 + x_4^2 &= c^2(t_1^2 + t_2^2 + t_3^2 + t_4^2) \\ &= c^2 \left\{ \left(\sum t_i \right)^2 - 2 \sum (t_i t_j) \right\} \\ &= c^2 \left\{ \left(\frac{\alpha}{c} \right)^2 - 0 \right\} = \alpha^2 \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad y_1^2 + y_2^2 + y_3^2 + y_4^2 &= \left(\sum y_i \right)^2 - 2 \sum (y_i y_j) \\ &= (\beta)^2 - 2c^2 \sum \left(\frac{1}{t_i t_j} \right) \\ &= (\beta)^2 - 2c^2 \left(\frac{\sum t_i t_j}{t_1 t_2 t_3 t_4} \right) \\ &= (\beta)^2 \end{aligned}$$

$$\text{(v)} \quad x_1 \cdot x_2 \cdot x_3 \cdot x_4 = c^4(t_1 t_2 t_3 t_4) = -c^4$$

$$\begin{aligned} \text{(vi)} \quad y_1 \cdot y_2 \cdot y_3 \cdot y_4 &= c^4 \left(\frac{1}{t_1 t_2 t_3 t_4} \right) \\ &= c^4 \left(\frac{1}{-1} \right) = -c^4 \end{aligned}$$

46. the equation of any normal to the given hyperbola at (x, y) is

$$\begin{aligned} \frac{x}{a^2}(k-y) &= \frac{y}{b^2}(x-b) \\ \Rightarrow \left(\frac{1}{a^2} + \frac{1}{b^2} \right) xy - \frac{y}{b} - \frac{kx}{a^2} &= 0 \\ \Rightarrow y &= \frac{b^2 k x}{a^2(e^2 x - b)} \end{aligned} \quad \dots \text{(i)}$$

The equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \dots \text{(ii)}$$

Solving Eqs (i) and (ii), we get,

$$\begin{aligned} \frac{x^2}{a^2} - \frac{b^4 k^2 x^2}{b^2 a^4 (e^2 x - b)^2} &= 1 \\ \Rightarrow a^2 e^4 x^4 - 2ba^2 e^2 x^3 - (a^2 b^2 + b^2 k^2 + a^4 e^4) x^2 + 2ba^4 e^2 x &+ a^4 b^2 - 0 \end{aligned} \quad \dots \text{(iii)}$$

Let x_1, x_2, x_3, x_4 are the roots of Eq. (iii).

$$\text{Then, } x_1 + x_2 + x_3 + x_4 = \frac{2b}{e^2},$$

$$\sum (x_i x_j) = -\frac{a^2 b^2 + b^2 k^2 + a^4 e^4}{a^2 e^4}$$

$$\sum (x_i x_j x_k) = -\frac{2ba^2}{e^2}$$

$$\text{and } \sum (x_i x_j x_k x_l) = \frac{a^2 b^2}{e^4}$$

$$\text{Therefore, } \sum \left(\frac{1}{x_i} \right) = \frac{\sum (x_i x_j x_k)}{x_1 x_2 x_3 x_4} = \frac{2e^2}{b}$$

$$\begin{aligned} \text{(i)} \quad (x_1 + x_2 + x_3 + x_4) &\left(\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4} \right) \\ &= \left(\sum x_i \right) \left(\sum \left(\frac{1}{x_i} \right) \right) = \frac{2b}{e^2} \times \frac{2e^2}{b} = 4 \end{aligned}$$

(ii) Similarly we can have

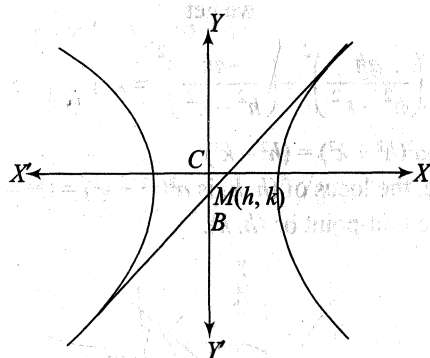
$$\begin{aligned} (y_1 + y_2 + y_3 + y_4) &\left(\frac{1}{y_1} + \frac{1}{y_2} + \frac{1}{y_3} + \frac{1}{y_4} \right) \\ &= \left(\sum y_i \right) \left(\sum \left(\frac{1}{y_i} \right) \right) \\ &= 4. \end{aligned}$$

47. The equation of the chord of contact of tangents drawn from the point $(2, 3)$ is

$$\begin{aligned} \frac{2x}{9} - \frac{y}{2} &= 1 \\ \Rightarrow 4x - 9y &= 18 \end{aligned}$$

48. Any tangent to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ is

$$\frac{x}{3} \sec \theta - \frac{y}{2} \tan \theta = 1 \quad \dots \text{(i)}$$



Let the tangent intersects the x -axis at A and y -axis at B , respectively.

Then $A = (3 \cos \theta, 0)$ and $B = (0, -2 \cot \theta)$

Let (h, k) be the mid-point of AB .

Therefore,

$$2h = 3 \cos \theta \text{ and } 2k = -2 \cot \theta$$

$$\Rightarrow \sec \theta = \frac{3}{2h} \text{ and } \tan \theta = -\frac{1}{k}$$

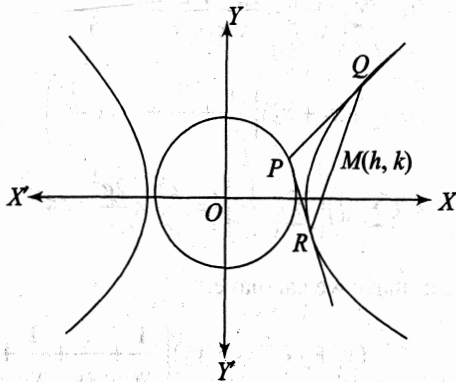
As we know that,

$$\sec^2 \theta - \tan^2 \theta = 1$$

$$\Rightarrow \frac{9}{4h^2} - \frac{1}{k^2} = 1$$

Hence, the locus of (h, k) is $\frac{9}{4x^2} - \frac{1}{y^2} = 1$.

49. Any point on the circle $x^2 + y^2 = a^2$ be $(a \cos \theta, a \sin \theta)$.



The equation of the chord of contact from the point $(a \cos \theta, a \sin \theta)$ to the hyperbola $x^2 - y^2 = a^2$ is

$$(a \cos \theta)x - (a \sin \theta)y = a^2$$

$$\Rightarrow x \cos \theta - y \sin \theta = a \quad \dots(i)$$

Let the mid-point be (h, k) .

The equation of the chord bisected at (h, k) to the hyperbola $x^2 - y^2 = a^2$ is

$$hx - ky = h^2 - k^2 \quad \dots(ii)$$

Since the Eqs (i) and (ii) are identical, so

$$\frac{h}{\cos \theta} = \frac{-k}{\sin \theta} = \frac{h^2 - k^2}{a}$$

$$\Rightarrow \cos \theta = \frac{ah}{h^2 - k^2} \text{ and } \sin \theta = \frac{-ak}{h^2 - k^2}$$

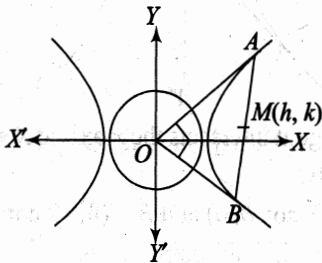
Squaring and adding, we get

$$\left(\frac{ah}{h^2 - k^2} \right)^2 + \left(\frac{-ak}{h^2 - k^2} \right)^2 = 1$$

$$\Rightarrow a^2(h^2 + k^2) = (h^2 - k^2)^2$$

Hence, the locus of (h, k) is $a^2(x^2 + y^2) = (x^2 - y^2)^2$.

50. Let the mid-point be (h, k) .



The equation of the chord bisected at (h, k) to the given hyperbola is

$$T = S_1$$

$$\Rightarrow \frac{hx}{a^2} - \frac{ky}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2} \quad \dots(i)$$

The equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \dots(ii)$$

Since the chord (i) subtends right angle at the centre, so we can write

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{\left(\frac{hx}{a^2} - \frac{ky}{b^2} \right)^2}{\left(\frac{h^2}{a^2} - \frac{k^2}{b^2} \right)^2}$$

$$\Rightarrow \frac{1}{a^2} \left(\frac{h^2}{a^2} - \frac{k^2}{b^2} \right)^2 x^2 - \frac{1}{b^2} \left(\frac{h^2}{a^2} - \frac{k^2}{b^2} \right)^2 y^2 = \frac{h^2}{a^4} x^2 + \frac{k^2}{b^4} y^2 - \frac{2hk}{a^2 b^2} xy \quad \dots(iii)$$

Equation (iii) will be a right angle, if co-efficient of $x^2 +$ co-efficient of $y^2 = 0$

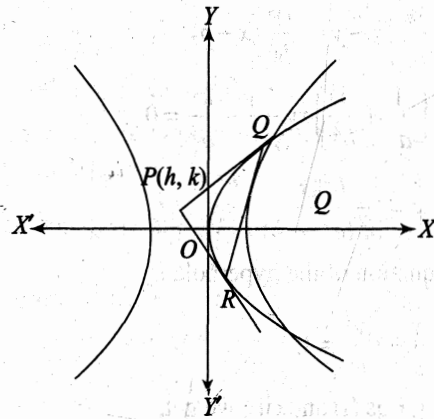
$$\Rightarrow \frac{1}{a^2} \left(\frac{h^2}{a^2} - \frac{k^2}{b^2} \right)^2 - \frac{h^2}{a^4} - \frac{1}{b^2} \left(\frac{h^2}{a^2} - \frac{k^2}{b^2} \right)^2 - \frac{k^2}{b^4} = 0$$

$$\Rightarrow \left(\frac{h^2}{a^2} - \frac{k^2}{b^2} \right)^2 \left(\frac{1}{a^2} - \frac{1}{b^2} \right) = \left(\frac{h^2}{a^4} + \frac{k^2}{b^4} \right)$$

Hence, the locus of (h, k) is

$$\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} \right)^2 \left(\frac{1}{a^2} - \frac{1}{b^2} \right) = \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} \right)$$

51. Let the point on the parabola be (h, k) .



The equation of the chord of contact of the parabola is

$$yk = 2a(x + h) \quad \dots(i)$$

$$\Rightarrow y = \frac{2a}{k}x + \frac{2ah}{k} \quad \dots(ii)$$

Since, the line (ii) is a tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \text{ so,}$$

$$c^2 = a^2 m^2 - b^2$$

$$\Rightarrow \left(\frac{2ah}{k} \right)^2 = a^2 \left(\frac{2a}{k} \right)^2 - b^2$$

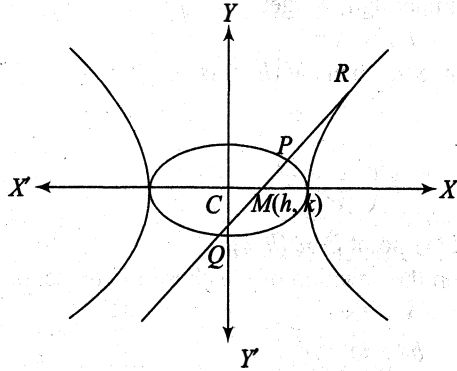
$$\Rightarrow 4a^2 h^2 = 4a^4 - k^2 b^2$$

Hence, the locus of (h, k) is

$$4a^2 x^2 = 4a^4 - y^2 b^2$$

52. Any tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $R(\theta)$ is

$$\frac{x}{a} \sec \theta - \frac{y}{b} \tan \theta = 1 \quad \dots(i)$$



Let the mid-point of PQ be (h, k) .

Then the equation of the chord bisected at (h, k) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

$$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} \quad \dots(ii)$$

Therefore, the Eqs (i) and (ii) are identical. So

$$\begin{aligned} \frac{\sec \theta / a}{\frac{h}{a^2}} &= \frac{-\tan \theta / b}{\frac{k}{b^2}} = \frac{1}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)} \\ \Rightarrow \frac{\sec \theta}{\frac{h}{a}} &= \frac{-\tan \theta}{\frac{k}{b}} = \frac{1}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)} \\ \Rightarrow \sec \theta &= \frac{\frac{h}{a}}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)} \text{ and } \tan \theta = \frac{\frac{k}{b}}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)} \end{aligned}$$

We know that,

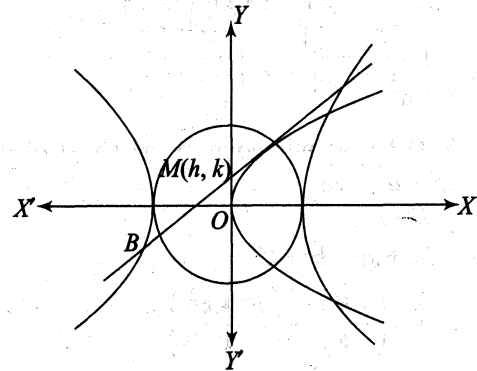
$$\begin{aligned} \sec^2 \theta - \tan^2 \theta &= 1 \\ \Rightarrow \frac{\left(\frac{h}{a}\right)^2}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2} - \frac{\left(\frac{k}{b}\right)^2}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2} &= 1 \\ \Rightarrow \left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right) &= \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2 \end{aligned}$$

Hence, the locus of (h, k) is $\left(\frac{x^2}{a^2} - \frac{y^2}{b^2}\right) = \left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2$.

53. If (h, k) be the mid-point of the chord of the hyperbola $x^2 - y^2 = a^2$, then
 $T = S_1$

$$\begin{aligned} \Rightarrow hx - ky &= h^2 - k^2 \\ \Rightarrow ky &= hx + (k^2 - h^2) \end{aligned}$$

$$\Rightarrow y = \left(\frac{h}{k}\right)x + \left(\frac{k^2 - h^2}{k}\right) \quad \dots(i)$$



If the line (i) touches the parabola $y^2 = 4ax$, so

$$\begin{aligned} c &= \frac{a}{m} \\ \Rightarrow \left(\frac{h^2 - k^2}{k}\right) &= \frac{a}{(h/k)} = \frac{ak}{h} \end{aligned}$$

$$\Rightarrow h(h^2 - k^2) = ak^2$$

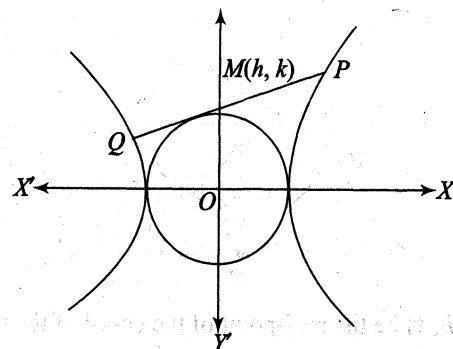
Hence, the locus of (h, k) is

$$x(x^2 - y^2) = ay^2$$

54. If (h, k) be the mid-point of the chord of the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \text{ then}$$

$$\begin{aligned} T &= S_1 \\ \Rightarrow \frac{hx}{a^2} - \frac{ky}{b^2} &= \frac{h^2}{a^2} - \frac{k^2}{b^2} \\ \Rightarrow \frac{ky}{b^2} &= \frac{hx}{a^2} - \left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right) \\ \Rightarrow y &= \left(\frac{b^2 h}{a^2 k}\right)x - \frac{b^2}{k} \left(\frac{h^2}{a^2} - \frac{k^2}{b^2}\right) \quad \dots(i) \end{aligned}$$



If the line (i) be a tangent to the circle $x^2 + y^2 = c^2$, then
 $C^2 = A^2(1 + m^2)$

$$\Rightarrow \frac{b^4 \left(\frac{h^2}{a^2} - \frac{k^2}{b^2} \right)^2}{k^2} = c^2 \left(1 + \frac{b^4 h^2}{a^4 k^2} \right)$$

$$\Rightarrow (b^2 h^2 - a^2 k^2)^2 = c^2 (a^4 k^2 + b^4 h^2)$$

Hence, the locus of (h, k) is

$$(b^2 x^2 - a^2 y^2)^2 = c^2 (a^4 y^2 + b^4 x^2)$$

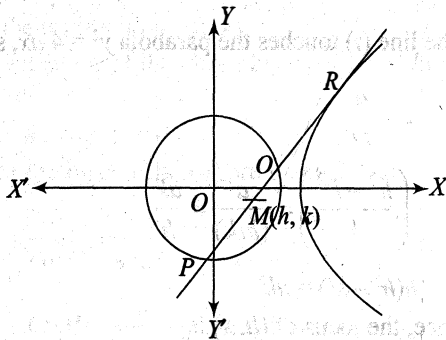
$$\Rightarrow \left(\frac{x^2}{a^2} - \frac{y^2}{b^2} \right)^2 = c^2 \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} \right)$$

55. If (h, k) be the mid-point of the chord of the circle $x^2 + y^2 = a^2$, then

$$T = S_1$$

$$\Rightarrow hx + ky = h^2 + k^2$$

$$\Rightarrow y = \left(-\frac{h}{k} \right) x + \left(\frac{h^2 + k^2}{k} \right) \quad (i)$$



If the line (i) be a tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, so,

$$c^2 = a^2 m^2 - b^2$$

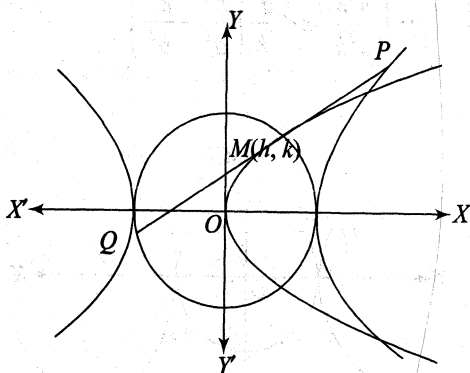
$$\Rightarrow \left(\frac{h^2 + k^2}{k} \right)^2 = a^2 \left(\frac{h^2}{k^2} \right) - b^2$$

$$\Rightarrow (h^2 + k^2)^2 = (a^2 h^2 - b^2 k^2)$$

Hence, the locus of (h, k) is

$$(x^2 + y^2)^2 = (a^2 x^2 - b^2 y^2)$$

56. Any tangent to the parabola $y^2 = 4ax$ at $(at^2, 2at)$ is $yt = x + at^2$... (i)



If (h, k) be the mid-point of the chord of the hyperbola $xy = c^2$, then

$$xk + yh = c^2 \quad \dots (ii)$$

Therefore, the Eqs (i) and (ii) are identical.

$$\text{Thus, } \frac{k}{1} = \frac{-h}{t} = \frac{-c^2}{at^2}$$

$$\Rightarrow t = \frac{-h}{k}, t^2 = \frac{-c^2}{ak}$$

Eliminating t , we get,

$$h^2 a = -k c^2$$

Hence the locus of (h, k) is

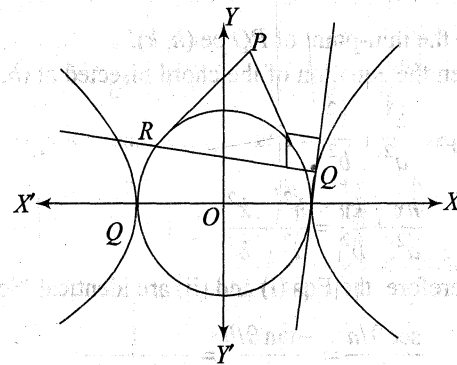
$$yc^2 + x^2 a = 0$$

$$\Rightarrow y = \left(-\frac{a}{c^2} \right) x^2.$$

57. Let the point P be (h, k) .

Then the equation of the chord of contact of the circle $x^2 + y^2 = a^2$ is

$$hx + ky = a^2 \quad \dots (i)$$



The equation of the normal chord of the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ at } (\phi) \text{ is}$$

$$ax \cos \phi - by \cot \phi = a^2 + b^2 \quad \dots (ii)$$

Equations (i) and (ii) are identical. Therefore,

$$\frac{a \cos \phi}{h} = -\frac{b \cot \phi}{k} = \frac{(a^2 + b^2)}{a^2}$$

$$\sec \phi = \frac{a^3}{h(a^2 + b^2)} \text{ and } \tan \phi = -\frac{a^2 b}{k(a^2 + b^2)}$$

We have,

$$\sec^2 \phi - \tan^2 \phi = 1$$

$$\Rightarrow \left(\frac{a^3}{h(a^2 + b^2)} \right)^2 - \left(-\frac{a^2 b}{k(a^2 + b^2)} \right)^2 = 1$$

$$\Rightarrow \left(\frac{a^6}{h^2} - \frac{a^4 b^2}{k^2} \right) = (a^2 + b^2)^2$$

$$\Rightarrow \left(\frac{a^2}{h^2} - \frac{b^2}{k^2} \right) = \left(\frac{a^2 + b^2}{a^2} \right)^2$$

Hence, the locus of (h, k) is

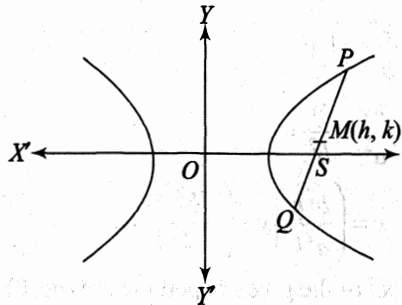
$$\left(\frac{a^2}{x^2} - \frac{b^2}{y^2} \right) = \left(\frac{a^2 + b^2}{a^2} \right)^2$$

58. If (h, k) be the mid-point of the chord of the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \text{ then}$$

$$T = S_1$$

$$\Rightarrow \frac{hx}{x^2} - \frac{ky}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2} \quad \dots(i)$$



which is passing through the focus $(ae, 0)$ of the given hyperbola.

Therefore,

$$\frac{aeh}{a^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$

$$\Rightarrow \frac{h^2}{a^2} - \frac{k^2}{b^2} = \frac{eh}{a}$$

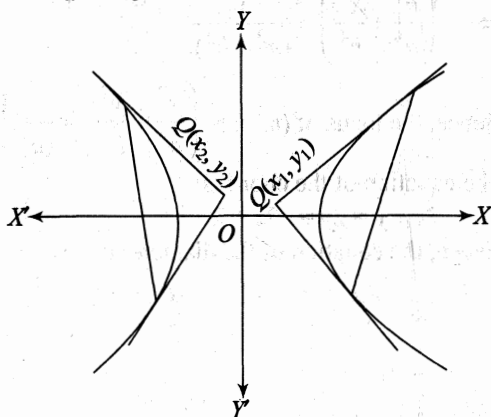
Hence, the locus of (h, k) is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \left(\frac{e}{a}\right)x$$

59. The equations of the chord of contact of the tangents to the given hyperbola at (x_1, y_1) and (x_2, y_2) are

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \quad \dots(i)$$

$$\text{and } \frac{xx_2}{a^2} - \frac{yy_2}{b^2} = 1 \quad \dots(ii)$$



The slopes of the lines (i) and (ii) are

$$m_1 = \left(\frac{b^2 x_1}{a^2 y_1}\right) \text{ and } m_2 = \left(\frac{b^2 x_2}{a^2 y_2}\right)$$

Since, (i) and (ii) meet at right angles, so

$$m_1 m_2 = -1$$

$$\Rightarrow \left(\frac{b^2 x_1}{a^2 y_1}\right) \times \left(\frac{b^2 x_2}{a^2 y_2}\right) = -1$$

$$\Rightarrow \frac{x_1 x_2}{y_1 y_2} = -\frac{a^4}{b^4}$$

Thus, $m = 4$ and $n = 4$

Hence, the value of $\left(\frac{m+n}{4}\right)^{10} = \left(\frac{4+4}{4}\right)^{10} = 2^{10} = 1024$.

60. The equation of the polar with respect to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

$$\Rightarrow \frac{x(-ae)}{a^2} - \frac{y \cdot 0}{b^2} = 1$$

$$\Rightarrow x = -\frac{a}{e}$$

61. The equation of the polars from points (x_1, y_1) and (x_2, y_2) to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ are}$$

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1 \quad \dots(i)$$

$$\text{and } \frac{xx_2}{a^2} - \frac{yy_2}{b^2} = 1 \quad \dots(ii)$$

Now, slopes of (i) and (ii) are

$$m_1 = \frac{b^2 x_1}{a^2 y_1} \text{ and } m_2 = \frac{b^2 x_2}{a^2 y_2}$$

Since the polars of the given points are perpendicular, so

$$m_1 \times m_2 = -1$$

$$\Rightarrow \left(\frac{b^2 x_1}{a^2 y_1}\right) \times \left(\frac{b^2 x_2}{a^2 y_2}\right) = -1$$

$$\Rightarrow \frac{x_1 x_2}{y_1 y_2} = -\frac{a^4}{b^4}$$

$$\Rightarrow \frac{x_1 x_2}{y_1 y_2} + \frac{a^4}{b^4} = 0$$

62. Let (x_1, y_1) be the pole of the hyperbola. Then the equation of the polar from a point (x_1, y_1) w.r.t. the hyperbola $x^2 - 3y^2 = 3$ is

$$xx_1 - 3yy_1 = 3 \quad \dots(i)$$

The equation of the given polar is

$$x - y = 3 \quad \dots(ii)$$

Therefore, the Eqs (i) and (ii) are identical. So

$$\frac{x_1}{1} = \frac{-3y_1}{-1} = \frac{3}{3}$$

$$\Rightarrow x_1 = 1, y_1 = \frac{1}{3}$$

Hence, the pole is $\left(1, \frac{1}{3}\right)$.

63. Let the pole be (h, k) .

The equation of the polar from the point (h, k) w.r.t. the

hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$\frac{hx}{a^2} - \frac{ky}{b^2} = 1 \quad \dots(i)$$

The equation of the normal chord of the given hyperbola at (ϕ) is

$$ax \cos \phi - by \cot \phi = (a^2 + b^2) \quad \dots(ii)$$

Equations (i) and (ii) are identical. Therefore,

$$\frac{a \cos \phi}{(h/a^2)} = \frac{b \cot \phi}{(k/b^2)} = \frac{(a^2 + b^2)}{1}$$

$$\Rightarrow \cos \phi = (a^2 + b^2) \frac{h}{a^3}, \cot \phi = (a^2 + b^2) \frac{k}{b^3}$$

$$\Rightarrow \sec \phi = \frac{a^3}{(a^2 + b^2)h}, \tan \phi = \frac{b^3}{(a^2 + b^2)k}$$

We know that,

$$\sec^2 \phi - \tan^2 \phi = 1$$

$$\Rightarrow \left(\frac{a^3}{(a^2 + b^2)h} \right)^2 - \left(\frac{b^3}{(a^2 + b^2)k} \right)^2 = 1$$

$$\Rightarrow \left(\frac{a^3}{h} \right)^2 - \left(\frac{b^3}{k} \right)^2 = (a^2 + b^2)^2$$

Hence, the locus of (h, k) is

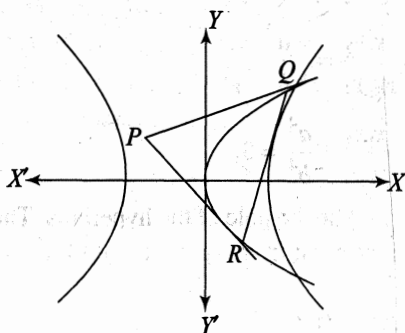
$$(a^6 y^2 - b^6 x^2) = (a^2 + b^2)^2 (x^2 y^2)$$

64. Let (h, k) be the pole.

The equation of the polar from the point (h, k) w.r.t. the parabola $y^2 = 4ax$ is

$$yk = 2a(x + h) = 2ax + 2ah$$

$$\Rightarrow y = \left(\frac{2a}{k} \right) x + \left(\frac{2ah}{k} \right) \quad \dots(i)$$



If the line (i) be a tangent to the hyperbola $x^2 - y^2 = a^2$, then

$$c^2 a^2 m^2 - a^2 = 1$$

$$\Rightarrow \left(\frac{2ah}{k} \right)^2 = a^2 \left(\frac{2a}{k} \right)^2 - a^2$$

$$\Rightarrow 4h^2 = 4a^2 - k^2$$

$$\Rightarrow h^2 + \frac{k^2}{4} = a^2$$

Hence, the locus of (h, k) is

$$x^2 + \frac{y^2}{4} = a^2$$

$$\Rightarrow 4x^2 + y^2 = 4a^2$$

65. Let (h, k) be the pole.

Then the equation of the polar w.r.t. the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{hx}{a^2} - \frac{ky}{b^2} = 1$$

$$\Rightarrow y = \left(\frac{b^2 h}{a^2 k} \right) x + \left(-\frac{b^2}{k} \right) \quad \dots(i)$$

The foci of the given hyperbola are $(ae, 0)$ and $(-ae, 0)$

The equation of the circle is

$$(x - ae)(x + ae) + y^2 = 0$$

$$\Rightarrow x^2 + y^2 = (ae)^2 \quad \dots(ii)$$

If the line (i) be a tangent to the circle (ii), then

$$c^2 = a^2(1 + m^2)$$

$$\Rightarrow \frac{b^4}{k^2} = (ae)^2 \left(1 + \frac{b^4 k^2}{a^4 h^2} \right)$$

$$\Rightarrow \frac{b^4}{k^2} = e^2 \left(\frac{a^4 h^2 + b^4 k^2}{a^2 k^2} \right)$$

$$\Rightarrow \frac{a^2 b^4}{e^2} = (a^4 h^2 + b^4 k^2)$$

$$\Rightarrow (a^4 h^2 + b^4 k^2) = \frac{a^4 b^4}{(a^2 + b^2)}$$

$$\Rightarrow \left(\frac{h^2}{a^4} + \frac{k^2}{b^4} \right) = \frac{1}{(a^2 + b^2)}$$

$$\text{Hence, the locus of } (h, k) \text{ is } \left(\frac{x^2}{a^4} + \frac{y^2}{b^4} \right) = \frac{1}{(a^2 + b^2)}$$

66. The equation of the chord is

$$7x + y = 2y = -7x + 2 \quad \dots(i)$$

Hence, the equation of the diameter is

$$y = \frac{b^2 x}{a^2 m} = \frac{7x}{3 \times (-7)} = -\frac{x}{3}$$

$$x + 3y = 0$$

67. The given line is $3x + 4y + 10 = 0$

$$y = \left(-\frac{3}{4} \right) x + \left(-\frac{5}{2} \right) \quad \dots(i)$$

Hence, the equation of the diameter corresponds to the line (i) is

$$y = \frac{b^2 x}{a^2 m} = \frac{4x}{9 \left(-\frac{3}{4} \right)} = -\frac{16}{27} x$$

$$\Rightarrow 16x + 27y = 0$$

68. The slope of the given chord is $m = \left(-\frac{2}{3}\right)$.

Hence, the equation of the diameter parallel to the given chord is

$$y = -\frac{b^2x}{a^2m} = -\frac{4x}{9\left(-\frac{2}{3}\right)} = -\frac{2}{3}x$$

$$\Rightarrow 2x + 3y = 0$$

69. Let the equation of the diameter, which is conjugate to $x = 2y$ is

$$y = m_1x.$$

As we know that two diameters $y = \left(\frac{1}{2}\right)x$ and $y = m_1x$ are conjugates, if

$$m_1m_2 = \frac{b^2}{a^2}$$

$$\Rightarrow m_1 \times \frac{1}{2} = \frac{16}{9}$$

$$\Rightarrow m_1 = \frac{32}{9}$$

Hence, the equation of the conjugate diameters is

$$y = \frac{32}{9}x$$

$$\Rightarrow 32x = 9y$$

70. Equations of the asymptotes to the hyperbola

$$xy - 2x - 3y = 0 \text{ is}$$

$$xy - 2x - 3y + \lambda = 0,$$

where λ is any constant such that it represents two straight lines.

Therefore,

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow 0 + 2 \times \left(\frac{-3}{2}\right) \times (-1) \times \left(\frac{1}{2}\right) - 0 - 0 - \lambda \left(\frac{1}{2}\right)^2 = 0$$

$$\Rightarrow \lambda = 6$$

Hence, the required asymptotes are

$$xy - 2x - 3y + 6 = 0$$

$$\Rightarrow (x-2)(y-3) = 0$$

$$\Rightarrow x = 2 \text{ and } y = 3$$

71. The equations of the asymptotes of the given curve is

$$3x^2 + 10xy + 8y^2 + 14x + 22y + \lambda = 0,$$

where λ is any constant such that it represents two straight lines.

Therefore,

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow 3 \cdot 8 \cdot \lambda - 2 \cdot 7 \cdot 11 \cdot 5 - 3 \cdot 121 - 8 \cdot 49 - \lambda \cdot 25 = 0$$

$$\Rightarrow 24\lambda - 770 - 363 - 392 - 25\lambda = 0$$

$$\Rightarrow \lambda = 15$$

Hence, the combined equation of the given asymptotes is

$$3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$$

72. Since the asymptotes are perpendicular to each other, so the hyperbola is rectangular. Hence, its eccentricity is $\sqrt{2}$.

73. The combined equation of the asymptotes is

$$(2x - y - 3)(3x + y - 7) = 0$$

$$\Rightarrow 6x^2 - xy - y^2 - 23x + 4y + 21 = 0$$

Let the equation of the hyperbola be

$$6x^2 - xy - y^2 - 23x + 4y + \lambda = 0,$$

where λ is any constant such that it represents two straight lines which passes through (1, 1), so $\lambda = 15$.

Hence, the equation of the hyperbola becomes

$$6x^2 - xy - y^2 - 23x + 4y + 15 = 0.$$

74. The combined equation of the asymptotes parallel to the lines $2x + 3y = 0$ and $3x + 2y = 0$ is

$$(2x + 3y + \lambda)(3x + 2y + \mu) = 0$$

which is passing through (1, 2).

Therefore,

$$(2 \cdot 1 + 3 \cdot 2 + \lambda)(3 \cdot 1 + 2 \cdot 2 + \mu) = 0$$

$$\Rightarrow (8 + \lambda)(7 + \mu) = 0$$

$$\Rightarrow \lambda = -8, \mu = -7$$

Thus, the combined equation of the asymptotes is

$$(2x + 3y - 8)(3x + 2y - 7) = 0$$

Let the equation of the hyperbola be

$$2x + 3y = 0 \text{ and } 2x - 3y = 0$$

which is passing through (5, 3). So

$$(2 \cdot 5 + 3 \cdot 3 - 8)(3 \cdot 5 + 2 \cdot 3 - 7) + \lambda = 0$$

$$\Rightarrow 11 \times 14 + \lambda = 0$$

$$\Rightarrow \lambda = -154$$

Hence, the equation of the hyperbola is

$$(2x + 3y - 8)(3x + 2y - 7) - 154 = 0.$$

75. The equation of the given hyperbola is

$$x^2 - 2y^2 = 2.$$

So, the equations of its asymptotes are $x - \sqrt{2}y = 0$ and $x + \sqrt{2}y = 0$.

Let any point on the hyperbola be $P(\sqrt{2} \sec \phi, \tan \phi)$.

Let PM and PN are two perpendiculars from the point P to the asymptotes.

Then, $PM \cdot PN$

$$= \left| \frac{\sqrt{2} \sec \phi - \sqrt{2} \tan \phi}{\sqrt{1+2}} \right| \times \left| \frac{\sqrt{2} \sec \phi + \sqrt{2} \tan \phi}{\sqrt{1+2}} \right|$$

$$= \frac{2}{3} \times (\sec \phi - \tan \phi) \times (\sec \phi + \tan \phi)$$

$$= \frac{2}{3} \times (\sec^2 \phi - \tan^2 \phi)$$

$$= \frac{2}{3}.$$

76. The equation of the given hyperbola is $\frac{x^2}{9} - \frac{y^2}{4} = 1$.

Thus, the equations of the asymptotes are

$$\left(\frac{x}{3} + \frac{y}{2}\right)\left(\frac{x}{3} - \frac{y}{2}\right) = 0$$

$$\Rightarrow \left(\frac{x}{3} + \frac{y}{2}\right) = 0 \text{ and } \left(\frac{x}{3} - \frac{y}{2}\right) = 0$$

$$\Rightarrow 2x + 3y = 0 \text{ and } 2x - 3y = 0$$

The equation of any tangent to the hyperbola

$$\frac{x^2}{9} - \frac{y^2}{4} = 1 \text{ is}$$

$$\frac{x}{3} \sec \phi - \frac{y}{2} \tan \phi = 1$$

Let the points of intersection of $2x + 3y = 0$, $2x - 3y = 0$

and $\frac{x}{3} \sec \phi - \frac{y}{2} \tan \phi = 1$ are O , P and Q respectively.

Therefore, $O = (0, 0)$,

$$P = \left(\frac{3}{\sec \phi + \tan \phi}, -\frac{2}{\sec \phi + \tan \phi} \right)$$

$$\text{and } Q = \left(\frac{3}{\sec \phi - \tan \phi}, \frac{2}{\sec \phi - \tan \phi} \right)$$

Hence, the area of ΔOPQ

$$\begin{aligned} &= \frac{1}{2} \begin{vmatrix} 0 & 0 \\ \frac{3}{\sec \phi + \tan \phi} & -\frac{2}{\sec \phi + \tan \phi} \\ \frac{3}{\sec \phi - \tan \phi} & \frac{2}{\sec \phi - \tan \phi} \end{vmatrix} \\ &= \frac{1}{2} (6 + 6) = 6 \text{ s. u.} \end{aligned}$$

77. Let the point P be $(a \sec \phi, b \tan \phi)$.

The equations of the asymptotes of the given hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ are}$$

$$bx - ay = 0 \text{ and } bx + ay = 0$$

Let PM and PN be two perpendiculars from the point P to the transverse axis and the asymptote

$$bx - ay = 0$$

Thus, $PM = b \tan \phi$

$$\text{and } PN = \frac{|ab \sec \phi - ab \tan \phi|}{\sqrt{b^2 + a^2}}$$

It is given that, $PM = PN$

$$\Rightarrow b \tan \phi = \frac{|ab \sec \phi - ab \tan \phi|}{\sqrt{b^2 + a^2}}$$

$$\Rightarrow \tan \phi = a \frac{\sec \phi - \tan \phi}{\sqrt{b^2 + a^2}}$$

78. Let the equation of the hyperbola be

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

...(i)

and any point on the hyperbola (i) be

$$P(a \sec \phi, b \tan \phi)$$

The equation of the tangent to the hyperbola (i) at P is

$$\frac{x}{a} \sec \phi - \frac{y}{b} \tan \phi = 1 \quad \dots(ii)$$

The equations of the asymptotes of the hyperbola (i) are

$$bx - ay = 0 \text{ and } bx + ay = 0$$

Let the points of the intersection of the asymptotes and the tangent are O , Q , R respectively.

Then, $O = (0, 0)$,

$$Q = [a(\sec \phi + \tan \phi), b(\sec \phi + \tan \phi)] \text{ and}$$

$$R = [a(\sec \phi - \tan \phi), b(\sec \phi - \tan \phi)]$$

Clearly, mid-point of QR is $(a \sec \phi, b \tan \phi)$, which is co-ordinates of P .

Thus, the area of ΔOQR

$$\begin{aligned} &= \frac{1}{2} \begin{vmatrix} 0 & 0 \\ a(\sec \phi + \tan \phi) & b(\sec \phi + \tan \phi) \\ a(\sec \phi - \tan \phi) & -b(\sec \phi - \tan \phi) \end{vmatrix} \\ &= \frac{1}{2} |-ab - ab| \\ &= ab. \end{aligned}$$

79. The equations of the asymptotes of the given hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ are}$$

$$bx - ay = 0 \text{ and } bx + ay = 0$$

Let P be any point on the given hyperbola be

$$(a \sec \phi, b \tan \phi).$$

It is given that, p_1 and p_2 be the lengths of perpendiculars from the point P to the asymptotes

$$bx - ay = 0 \text{ and } bx + ay = 0$$

$$\text{Thus, } p_1 = \frac{|ab \sec \phi - ab \tan \phi|}{\sqrt{b^2 + a^2}}$$

$$\text{and } p_2 = \frac{|ab \sec \phi + ab \tan \phi|}{\sqrt{b^2 + a^2}}$$

Therefore, $\frac{1}{p_1 p_2}$

$$\begin{aligned} &= \frac{a^2 + b^2}{(ab \sec \phi - ab \tan \phi)(ab \sec \phi + ab \tan \phi)} \\ &= \left(\frac{a^2 + b^2}{a^2 b^2} \right) = \frac{1}{a^2} + \frac{1}{b^2} \end{aligned}$$

Hence, the result.

80. The equation of the normal at $\left(ct_1, \frac{c}{t_1} \right)$ is

$$t_1^3 x - t_1 y - ct_1^4 + c = 0 \quad \dots(i)$$

Also it meets the hyperbola again at

$$\left(ct_2, \frac{c}{t_2} \right)$$

Therefore,

$$ct_2t_1^3 - c\frac{t_1}{t_2} - ct_1^4 + c = 0$$

$$\Rightarrow t_2t_1^3 - \frac{t_1}{t_2} - t_1^4 + 1 = 0$$

$$\Rightarrow t_2^2t_1^3 - t_1 - t_1^4t_2 + t_2 = 0$$

$$\Rightarrow t_2t_1^3(t_2 - t_1) + (t_2 - t_1) = 0$$

$$\Rightarrow (t_2t_1^3 + 1)(t_2 - t_1) = 0$$

$$\Rightarrow (t_2t_1^3 + 1) = 0 \quad (\because t_1 \neq t_2)$$

$$\Rightarrow t_2t_1^3 = -1$$

Hence, the result.

81. Let P , Q and R are the vertices of a triangle such that

$$P = \left(ct_1, \frac{c}{t_1}\right), Q = \left(ct_2, \frac{c}{t_2}\right), R = \left(ct_3, \frac{c}{t_3}\right)$$

$$\text{Now, slope of } QR = \frac{\frac{c}{t_3} - \frac{c}{t_2}}{ct_3 - ct_2} = -\frac{1}{t_2t_3}$$

Therefore, slope of PM is t_2t_3 .

The equation of the perpendicular PM on QR is

$$y - \frac{c}{t_1} = t_2t_3(x - ct_1) \quad \dots(i)$$

Similarly, the equation of the perpendicular BN on PR is

$$y - \frac{c}{t_2} = t_1t_3(x - ct_2) \quad \dots(ii)$$

Solving Eq. (i) and (ii), we get,

$$x = -\frac{c}{t_1t_2t_3} \text{ and } y = -ct_1t_2t_3.$$

Thus, the point $\left(-\frac{c}{t_1t_2t_3}, -ct_1t_2t_3\right)$ lies on the rectangular hyperbola $xy = c^2$.

Hence, the result.

82. The equation of the normal to the rectangular hyperbola $xy = c^2$ at t is

$$xt^2 - y = ct^3 - \frac{c}{t} \quad \dots(i)$$

Let the pole be (h, k) .

Then the equation of the polar from the point (h, k) to the rectangular hyperbola

$$xy = c^2 \text{ is}$$

$$xk + yh = 2c^2 \quad (ii)$$

Therefore, the Eqs (i) and (ii) are identical. So

$$\frac{k}{t^2} = \frac{h}{-1} = \frac{2c^2}{ct^3 - \frac{c}{t}}$$

$$\Rightarrow t^2 = -\frac{k}{h} \text{ and } h = -\frac{2ct}{t^4 - 1}$$

Eliminating t , we get

$$(h^2 - k^2)^2 + 4c^2hk = 0$$

Hence, the locus of (h, k) is

$$(x^2 - y^2)^2 + 4c^2xy = 0$$

83. The equations of the asymptotes of the rectangular hyperbola $xy = c^2$ are $x = 0$ and $y = 0$.

Clearly, the angle between the asymptotes

$$= \frac{\pi}{2}$$

$$\Rightarrow 2\alpha = \frac{\pi}{2}$$

$$\Rightarrow \alpha = \frac{\pi}{4}$$

Thus, the eccentricity,

$$e = \sqrt{2} = \sec\left(\frac{\pi}{4}\right) = \sec\alpha$$

84. Let the equation of the circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \dots(i)$$

and the equation of the given hyperbola be

$$xy = 1 \quad \dots(ii)$$

Solving, we get

$$x^2 + \frac{1}{x^2} + 2gx + \frac{2f}{x} + c = 0$$

$$\Rightarrow x^4 + 2gx^3 + cx^2 + 2fx + 1 = 0$$

Let its roots are x_1, x_2, x_3 and x_4 .

Then, $x_1x_2x_3x_4 = 1$

Similarly, we can easily prove that

$$y_1y_2y_3y_4 = 1$$

85. Any point on the rectangular hyperbola

$$xy = c^2 \text{ is } P\left(ct, \frac{c}{t}\right)$$

The equation of any tangent to the rectangular hyperbola

$xy = c^2$ at t is

$$\frac{x}{t} + yt = 2c \quad \dots(i)$$

The equation of any normal to the rectangular hyperbola

$xy = c^2$ at t is

$$xt^3 - yt - xt^4 + c = 0 \quad \dots(ii)$$

Therefore, $a_1 = 2ct, a_2 = \frac{2c}{t}$

$$\text{and } b_1 = c\left(t - \frac{1}{t^3}\right), b_2 = c\left(\frac{1}{t} - t^3\right)$$

Now,

$$\begin{aligned} a_1a_2 + b_1b_2 &= 2c^2t\left(t - \frac{1}{t^3}\right) + \frac{2c^2}{t}\left(\frac{1}{t} - t^3\right) \\ &= 2c^2\left(t^2 - \frac{1}{t^2} + \frac{1}{t^2} - t^2\right) = 0 \end{aligned}$$

86. We have, $e_1 = \sqrt{2}$ and $e_2 = \sqrt{2}$

$$\text{Now, } (e_1 + e_2)^2 = (\sqrt{2} + \sqrt{2})^2 = (2\sqrt{2})^2 = 8$$

87. Any point on the given hyperbola

$$\frac{x^2}{2} - y^2 = 1 \text{ be } P(\sqrt{2} \sec \phi, \tan \phi).$$

The equations of the asymptotes of the given hyperbola are

$$x - \sqrt{2}y = 0 \text{ and } x + \sqrt{2}y = 0.$$

Let PM and PN be the lengths of perpendiculars from the point P on the asymptotes.

Thus,

$$PM \cdot PN$$

$$= \left| \frac{\sqrt{2} \sec \phi - \sqrt{2} \tan \phi}{\sqrt{1+2}} \right| \times \left| \frac{\sqrt{2} \sec \phi + \sqrt{2} \tan \phi}{\sqrt{1+2}} \right|$$

$$= \frac{2}{3}$$

$$= (1 \pm 2\sqrt{2}, 2)$$

88. Let the points A, B, C be

$$\left(ct_1, \frac{c}{t_1} \right), \left(ct_2, \frac{c}{t_2} \right), \left(ct_3, \frac{c}{t_3} \right) \text{ respectively.}$$

(i) Then the area of the ΔABC

$$= \frac{1}{2} \begin{vmatrix} ct_1 & \frac{c}{t_1} & 1 \\ ct_2 & \frac{c}{t_2} & 1 \\ ct_3 & \frac{c}{t_3} & 1 \end{vmatrix}$$

$$= \frac{1}{2} \frac{c^2}{t_1 t_2 t_3} \begin{vmatrix} t_1^2 & 1 & t_1 \\ t_2^2 & 1 & t_2 \\ t_3^2 & 1 & t_3 \end{vmatrix}$$

$$= \frac{c^2}{2} \times \frac{(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)}{t_1 t_2 t_3}$$

(ii) The equations of the tangents at A, B and C are

$$\frac{x}{t_1} + yt_1 = 2c \quad \dots(i)$$

$$\frac{x}{t_2} + yt_2 = 2c \quad \dots(ii)$$

$$\text{and } \frac{x}{t_3} + yt_3 = 2c \quad \dots(iii)$$

Thus, the points of intersections of (i) and (ii), (i) and (iii), (ii) and (iii) meet at P, Q, R respectively.

$$\text{Thus, } P = \left(\frac{2ct_1 t_2}{t_1 + t_2}, \frac{2c}{t_1 + t_2} \right),$$

$$Q = \left(\frac{2ct_1 t_3}{t_1 + t_3}, \frac{2c}{t_1 + t_3} \right)$$

$$\text{and } R = \left(\frac{2ct_2 t_3}{t_2 + t_3}, \frac{2c}{t_2 + t_3} \right) \text{ respectively}$$

Hence, the area of the ΔPQR

$$= \frac{1}{2} \begin{vmatrix} \frac{2ct_1 t_2}{t_1 + t_2} & \frac{2c}{t_1 + t_2} & 1 \\ \frac{2ct_1 t_3}{t_1 + t_3} & \frac{2c}{t_1 + t_3} & 1 \\ \frac{2ct_2 t_3}{t_2 + t_3} & \frac{2c}{t_2 + t_3} & 1 \end{vmatrix}$$

$$= \frac{2c^2 (t_1 - t_2)(t_2 - t_3)(t_3 - t_1)}{(t_1 + t_2)(t_1 + t_3)(t_2 + t_3)}$$

89. The given rectangular hyperbola is

$$xy = 18$$

...(i)

Replacing x by $x \cos(45^\circ) + y \sin(45^\circ)$ and y by $-x \sin(45^\circ) + y \cos(45^\circ)$ in (i), we get

$$\left(\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} \right) \left(-\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} \right) = 18$$

$$\Rightarrow x^2 - y^2 = -18$$

Hence, the length of the transverse axis

$$= 2a = 2.6 = 12 \text{ units.}$$

90. Let (h, k) be any point.

The equation of the chord of contact of the tangents from (h, k) to the circle $x^2 + y^2 = 4$ is

$$hx + ky = 4.$$

Also, the given hyperbola is

$$xy = 1$$

$$\Rightarrow x \left(\frac{4 - hx}{k} \right) = 1$$

$$\Rightarrow 4x - hx^2 = k$$

$$\Rightarrow hx^2 - 4x + k = 0$$

Thus, its roots are equal. So

$$D = 0$$

$$\Rightarrow 16 - 4hk = 0$$

$$\Rightarrow hk = 4$$

Hence, the locus of (h, k) is $xy = 4$.

91. The combined equation of the asymptotes of the given hyperbola is

$$xy - hx - ky + \lambda = 0$$

where λ is any constant such that it represents two straight lines.

Therefore,

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0.$$

$$\Rightarrow 0 + 2 \left(-\frac{k}{2} \right) \left(-\frac{h}{2} \right) \left(\frac{1}{2} \right) - \frac{\lambda}{4} = 0$$

$$\Rightarrow \lambda = hk$$